Real Analysis and Measure Theory

McGill MATH 564

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Written up from Professor Toth's handwritten lecture notes: Section n covers Lecture n, and Lectures 7–9 include their second parts. Slips in the original notes have been corrected, and routine steps left as exercises are filled in. Paragraphs labelled Explanation. are additions and are not from the lectures.

Definitions **Results** **Examples**

remarks, notes and explanations run inline, with an underlined label

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1 Course Overview and the Riemann Integral

Question. What is measure theory, and why should we care?

Overview. Let $E \subseteq \mathbb{R}^n$ be a bounded domain whose boundary ∂E is piecewise smooth. Call such sets *regular*. For regular E the Riemann integral gives the volume

$$\mathrm{vol}(E) = |E| = \underbrace{\int_E dx_1\, dx_2 \cdots dx_n}_{\text{Riemann integral}}.$$

Taking limits in analysis can replace regular sets (and functions) by much more singular ones, and the Riemann integral may not apply to the limits. So we want to extend $|E|$ to more general sets $E \subseteq \mathbb{R}^n$. The extension is denoted $m(E)$, the *Lebesgue measure* of E, and it should satisfy:

① E regular $\implies m(E) = |E|$.
② m is monotone non-decreasing: $E' \subseteq E \implies m(E') \leq m(E)$.
③ m is countably additive (Carathéodory construction).

The same question arises for functions: which $f : E \to \mathbb{R}$ can be integrated? For the Riemann integral, $\int_E f(x)\, dx$ is well-defined when E is regular and $f \in C^0(\overline{E})$. We want to integrate a broader class of functions, the *measurable functions*, which may be singular but behave well under limits. The extended (Lebesgue) integral satisfies

$$\underbrace{\int_E f\, dm}_{\text{Lebesgue}} = \underbrace{\int_E f(x)\, dx}_{\text{Riemann}} \qquad \text{for } E \subseteq \mathbb{R}^n \text{ regular and } f \in C^0(\overline{E}), \tag{$*$}$$

but the left side of $(*)$ is defined for a much larger class of f's and E's, and

$$\lim_{n\to\infty} \int_E f_n\, dm = \int_E \lim_{n\to\infty} f_n\, dm$$

holds for a large class of sequences $\{f_n\}$. In fact, the class of measurable functions is closed under pointwise limits.

Explanation. Countable additivity says: if a set is cut into countably many disjoint pieces, its measure is the sum of the measures of the pieces. This is what makes limits work, and it is where the Riemann volume $|E|$ breaks down: $\mathbb{Q} \cap [0,1]$ is a countable union of points, each of volume 0, yet it has no Riemann volume, since its indicator function is not Riemann integrable (Example 2 below).

1.1 Review of Riemann Integration

Definition

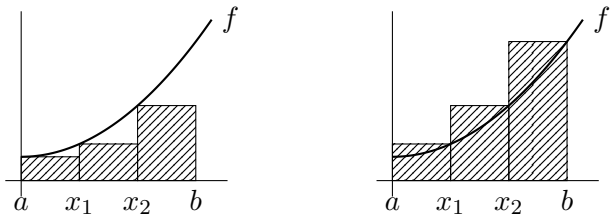
A *partition* of $[a,b]$ is a finite set $P = \{a = x_0 < x_1 < \cdots < x_{n-1} < x_n = b\}$.

Definition

Given $f : [a, b] \to \mathbb{R}$ bounded and a partition P, the *lower* and *upper Riemann sums* are

$$L(f, P) := \sum_{j=1}^{n} (x_j - x_{j-1}) \inf_{[x_{j-1}, x_j]} f,$$

$$U(f, P) := \sum_{j=1}^{n} (x_j - x_{j-1}) \sup_{[x_{j-1}, x_j]} f.$$



$L(f, P)$ $U(f, P)$

The key fact: if $P' \supseteq P$ is a *refinement* (e.g. $P' = P \cup \{y_1, \dots, y_m\}$), then

$$L(f, P) \le L(f, P') \le U(f, P') \le U(f, P). \tag{†}$$

Explanation. (†) holds because refining only splits intervals, and on a smaller interval the infimum can only go up and the supremum can only go down; the middle inequality is just inf $\le$ sup.

Definition

For $f : [a, b] \to \mathbb{R}$ bounded, the *lower* and *upper Riemann integrals* are

$$L(f) = \underline{\int_a^b} f\,dx := \sup_P L(f, P), \qquad U(f) = \overline{\int_a^b} f\,dx := \inf_P U(f, P).$$

Clearly $L(f) \le U(f)$: for any partitions P, Q, (†) for the common refinement $P \cup Q$ gives $L(f, P) \le L(f, P \cup Q) \le U(f, P \cup Q) \le U(f, Q)$; take $\sup_P$, then $\inf_Q$.

Definition

A bounded $f : [a, b] \to \mathbb{R}$ is *Riemann integrable*, $f \in \mathrm{RI}([a, b])$, if $U(f) = L(f)$. In that case the *Riemann integral* is

$$\int_a^b f(x)\,dx := U(f) = L(f).$$

Theorem

If $f \in C^0([a, b])$, then $f \in \mathrm{RI}([a, b])$.

Proof

Step 1. For any $\varepsilon > 0$ there is a partition P_ε with $U(f, P_\varepsilon) - L(f, P_\varepsilon) < \varepsilon$. Indeed, f is uniformly

continuous on $[a,b]$, so there is $\delta > 0$ with $|f(x) - f(y)| < \varepsilon/(b-a)$ whenever $|x-y| < \delta$. Take P_ε with all subintervals shorter than δ. On each subinterval $\sup f - \inf f < \varepsilon/(b-a)$, since both are attained, and summing gives $U(f,P_\varepsilon) - L(f,P_\varepsilon) < \varepsilon$.

Step 2. By monotonicity,

$$L(f,P_\varepsilon) \le L(f) \le U(f) \le U(f,P_\varepsilon),$$

so with Step 1, $0 \le U(f) - L(f) < \varepsilon$. Since $\varepsilon > 0$ is arbitrary, $U(f) = L(f)$.

2 Deficiencies of Riemann Integration and Set Theory

2.1 Deficiencies of Riemann Integration

Example (1)

Let $f(x) = 1/\sqrt{x}$ for $0 < x \le 1$ and $f(0) = 0$. Then $f \notin \mathrm{RI}([0,1])$: for *any* partition P of $[0,1]$, $\sup_{[x_0,x_1]} f = +\infty$, so

$$U(f,P) = \sum_{j=1}^{n} (x_j - x_{j-1}) \sup_{[x_{j-1},x_j]} f = +\infty.$$

In calculus one instead *defines* the improper integral

$$\int_0^1 f\,dx := \lim_{a \searrow 0} \int_a^1 \frac{dx}{\sqrt{x}} = \lim_{a \searrow 0} \left(2 - 2\sqrt{a}\right) = 2.$$

This is not satisfactory: f itself is still not integrable. Lebesgue integration solves this issue.

Example (2)

Let $\{r_k\}_{k=1}^{\infty}$ be an enumeration of $\mathbb{Q} \cap [0,1]$ and

$$f_k(x) = \begin{cases} 1, & x \in \{r_1, \dots, r_k\}, \\ 0, & \text{otherwise.} \end{cases}$$

For each finite k, $f_k \in \mathrm{RI}([0,1])$ and $\int_0^1 f_k(x)\,dx = 0$: every subinterval contains points where $f_k = 0$, so $L(f_k,P) = 0$; and at most $2k$ subintervals contain one of $r_1, \dots, r_k$, so $U(f_k,P) \le 2k \max_j (x_j - x_{j-1})$, which is as small as we like.

Question. Is $\lim_{k\to\infty} \int_0^1 f_k(x)\,dx = \int_0^1 \lim_{k\to\infty} f_k(x)\,dx$?

Answer. No; the right side does not even make sense. The pointwise limit is

$$f(x) = \lim_{k\to\infty} f_k(x) = \begin{cases} 1, & x \in \mathbb{Q} \cap [0,1], \\ 0, & \text{otherwise,} \end{cases}$$

and $f \notin \mathrm{RI}([0,1])$: every interval $[x_{j-1}, x_j]$ contains both rationals and irrationals ($\mathbb{Q}$ and $\mathbb{R} \setminus \mathbb{Q}$ are dense in $\mathbb{R}$), so for every partition P,

$$U(f,P) = 1 \quad \text{and} \quad L(f,P) = 0, \qquad \text{hence} \quad U(f) = 1 \neq 0 = L(f).$$

Upshot. The Riemann integral does not, in general, behave well with respect to taking limits, and it cannot integrate unbounded functions such as $1/\sqrt{x}$ on $(0,1]$.

Explanation. With the Lebesgue integral the answer to the question above becomes yes: $\mathbb{Q} \cap [0,1]$ is countable, so it has measure zero and $\int f\, dm = 0 = \lim_k \int_0^1 f_k\, dx$. Since the f_k increase to f, this exchange of limit and integral is an instance of the monotone convergence theorem (Section 12).

2.2 Notation

For more details see Folland, Chapter 0. Let X be a set.

- $\mathcal{P}(X) = 2^X = \{E \mid E \subseteq X\}$, the *power set* of X.
- For $E, F \subseteq X$: $E \setminus F = E - F = \{x \in E \mid x \notin F\}$.
- *Symmetric difference*: $E \triangle F := (E \setminus F) \cup (F \setminus E)$.
- *Complement*: $E^c := X \setminus E$.
- For a map $f : X \to Y$, $E \subseteq Y$ and $D \subseteq X$:

$$f^{-1}(E) := \{x \in X \mid f(x) \in E\}, \qquad f(D) := \{f(x) \mid x \in D\}.$$

Proposition (Basic relations)

Let $f : X \to Y$, let $\{E_\alpha\}_{\alpha \in A}$ be any family of subsets of Y, and let $E \subseteq Y$. Then

① $f^{-1}\big(\bigcup_{\alpha\in A} E_\alpha\big) = \bigcup_{\alpha\in A} f^{-1}(E_\alpha)$,
② $f^{-1}\big(\bigcap_{\alpha\in A} E_\alpha\big) = \bigcap_{\alpha\in A} f^{-1}(E_\alpha)$,
③ $f^{-1}(E^c) = \big(f^{-1}(E)\big)^c$ (complements in Y and in X).

Proof

Unwind the definitions. For ①, $x \in f^{-1}\big(\bigcup_\alpha E_\alpha\big) \iff f(x) \in E_\alpha$ for some $\alpha \iff x \in \bigcup_\alpha f^{-1}(E_\alpha)$; ② is the same with "for all α"; and for ③, $x \in f^{-1}(E^c) \iff f(x) \notin E \iff x \notin f^{-1}(E)$.

Explanation. No assumption such as $E_\alpha \subseteq \operatorname{Ran} f$ is needed. Images behave worse: $f\big(\bigcup_\alpha D_\alpha\big) = \bigcup_\alpha f(D_\alpha)$ still holds, but only $f(D_1 \cap D_2) \subseteq f(D_1) \cap f(D_2)$, with strict inclusion e.g. when f is constant and D_1, D_2 are disjoint and nonempty.

2.3 Equivalence Relations

Definition

An *equivalence relation* on X is a subset $R \subseteq X \times X$, written $xRy :\iff (x,y) \in R$, such that

ⓐ (Reflexivity) xRx for all $x \in X$;
ⓑ (Symmetry) $xRy \iff yRx$ for all $x, y \in X$;
ⓒ (Transitivity) xRy and $yRz \implies xRz$, for all $x, y, z \in X$.

Explanation. An equivalence relation is the same thing as a partition of X into disjoint *classes* $[x] = \{y \in X \mid yRx\}$: two points are related exactly when they lie in the same class. For example, $x \sim y \iff x - y \in \mathbb{Q}$ splits $[0,1)$ into countable classes.

2.4 Orderings

Definition

A *partial ordering* on $X \neq \emptyset$ is a relation R on X such that

(i) xRy and $yRz \implies xRz$;
(ii) xRy and $yRx \implies x = y$;
(iii) xRx for all x.

We denote a partial ordering by "$\leq$". A set with a partial ordering is a *partially ordered set*, or *poset*.

Definition

A partial ordering $\leq$ on X is a *total* (*linear*) *ordering* if for all $x, y \in X$, either $x \leq y$ or $y \leq x$. (E.g. $(\mathbb{N}, \leq)$.)

Definition

Let $(X, \leq)$ be a poset. A *maximal* (resp. *minimal*) element is an $x \in X$ such that the only $y \in X$ with $x \leq y$ (resp. $y \leq x$) is $y = x$.

Explanation. Maximal means that nothing lies strictly above x, not that x lies above everything. Take the subsets of $\{1,2,3\}$ with at most two elements, ordered by $\subseteq$. The sets $\{1,2\}$, $\{1,3\}$ and $\{2,3\}$ are all maximal, and none of them is the largest. In a total ordering, maximal and largest are the same.

Definition

A totally ordered set $(X, \leq)$ is *well-ordered* if every nonempty $E \subseteq X$ has a minimal element (which is then its least element).

Example

$(\mathbb{N}, \leq)$ is well-ordered.

Definition

Let $(X, \leq)$ be a poset. A totally ordered subset $S \subseteq X$ is also called a *chain*. A chain S is *maximal* if for every chain S' with $S \subseteq S' \subseteq X$ we have $S' = S$.

2.5 The Hausdorff Maximal Principle and Zorn's Lemma

Hausdorff Maximal Principle

Every poset $(X, \leq)$ has a maximal totally ordered subset (a maximal chain).

Zorn's Lemma

Let $(X, \leq)$ be a poset. If every totally ordered subset of X has an upper bound in X, then X has a maximal element.

Explanation. An upper bound of a chain C is any $b \in X$ with $c \leq b$ for all $c \in C$. It need not lie in C, so it is not the same as a maximal element of C: in $[0,1]$ with the usual order, the chain $[0,1)$ has the upper bound 1 but no maximal element. In applications (a maximal partial choice function in the theorem below, a basis of a vector space, a maximal ideal), the upper bound of a chain of partial objects is usually their union.

2.6 Axiom of Choice

Definition

Let A be an index set and $\{X_\alpha\}_{\alpha \in A}$ a collection of nonempty sets. The *product set* is

$$X = \prod_{\alpha \in A} X_\alpha := \left\{ f : A \to \bigcup_{\alpha \in A} X_\alpha \;\middle|\; f(\alpha) \in X_\alpha \text{ for all } \alpha \in A \right\}.$$

Remark. This definition is needed to deal with *uncountable* index sets A. For countable (e.g. finite) A it can be identified with the usual Cartesian product, e.g.

$$\prod_{j=1}^{2} X_j = \{f : \{1,2\} \to X_1 \cup X_2 \mid f(1) \in X_1,\ f(2) \in X_2\} \cong X_1 \times X_2, \qquad f \mapsto (f(1), f(2)).$$

There are natural projections $\pi_\alpha : \prod_{\beta \in A} X_\beta \to X_\alpha$, $\pi_\alpha(f) = f(\alpha)$.

Axiom of Choice

Let $\{X_\alpha\}_{\alpha \in A}$ be a collection of nonempty sets. Then $\prod_{\alpha \in A} X_\alpha \neq \emptyset$, i.e. there is a *choice function*: a function $f : A \to \bigcup_{\alpha \in A} X_\alpha$ with $f(\alpha) \in X_\alpha$ for every $\alpha \in A$.

Explanation. For finitely many sets, $\prod X_\alpha \neq \emptyset$ can be proved by induction without the axiom. The axiom is what allows infinitely many choices with no rule for making them, e.g. one point from each class of $x \sim y \iff x - y \in \mathbb{Q}$ on $[0,1)$; Section 3 uses such a choice to build a non-measurable set.

Theorem

In ZF set theory: Hausdorff Maximal Principle $\iff$ Zorn's Lemma $\iff$ Axiom of Choice.

Proof

HMP $\Leftrightarrow$ Zorn was an exercise in the lecture. AoC $\Rightarrow$ Zorn is much more involved and is left out.

(HMP $\Rightarrow$ Zorn) Assume HMP, and let $(X, \leq)$ be a poset in which every chain has an upper bound. Let C be a maximal chain in X, and let $b \in X$ be an upper bound of C. Then b is maximal in X. Otherwise there is $b' \in X$ with $b < b'$ (i.e. $b \leq b'$ and $b \neq b'$); then $b' \notin C$ (every $c \in C$ satisfies $c \leq b < b'$), and $C \cup \{b'\}$ is a chain strictly containing C, contradicting the maximality of C.

(Zorn $\Rightarrow$ HMP) Let $\mathcal{C}_X$ be the set of all chains in X, partially ordered by inclusion; $\mathcal{C}_X \neq \emptyset$, since every singleton $\{x\}$ is a chain. Let γ be a chain in $(\mathcal{C}_X, \subseteq)$. If $\gamma = \emptyset$, any element of $\mathcal{C}_X$ is an upper bound. Otherwise $\bigcup_{C \in \gamma} C$ is again a chain in X (any two of its elements lie in a common $C \in \gamma$, since γ is totally ordered by inclusion), and it is an upper bound for γ. By Zorn's lemma, $(\mathcal{C}_X, \subseteq)$ has a maximal element T, which by definition is a maximal chain in $(X, \leq)$.

(Zorn $\Rightarrow$ AoC) Let $\{X_\alpha\}_{\alpha \in A}$ be a collection of nonempty sets, and let $\mathcal{F}$ be the family of partial choice functions,

$$\mathcal{F} = \left\{ f : S \to \bigcup_{\alpha \in A} X_\alpha \;\middle|\; S \subseteq A,\ f(\alpha) \in X_\alpha \text{ for all } \alpha \in S \right\}.$$

$\mathcal{F} \neq \emptyset$, since it contains the empty function ($S = \emptyset$). For $f_1 : S_1 \to \bigcup X_\alpha$ and $f_2 : S_2 \to \bigcup X_\alpha$ in $\mathcal{F}$, say $f_1 \leq f_2$ if $S_1 \subseteq S_2$ and $f_2|_{S_1} = f_1$, i.e. f_2 extends f_1. Viewing functions as sets of pairs, this says $f_1 \subseteq f_2$, so $\leq$ is a partial ordering. For a chain $\{f_i : S_i \to \bigcup X_\alpha\}_{i \in I}$ in $\mathcal{F}$, define g on $\bigcup_{i \in I} S_i$ by $g(\alpha) := f_i(\alpha)$ for $\alpha \in S_i$. This is well-defined: if $\alpha \in S_i \cap S_j$, one of f_i, f_j extends the other, so $f_i(\alpha) = f_j(\alpha)$. (As a set of pairs, $g = \bigcup_{i \in I} f_i$; for the empty chain, g is the empty function.) Then $g(\alpha) \in X_\alpha$ for all α and $f_i \leq g$ for all i, so $g \in \mathcal{F}$ is an upper bound of the chain. By Zorn's lemma, $(\mathcal{F}, \leq)$ has a maximal element $f : S \to \bigcup X_\alpha$. If $S \neq A$, pick $\alpha_0 \in A \setminus S$ and $a \in X_{\alpha_0}$. Since $\alpha_0 \notin S$, $f \cup \{(\alpha_0, a)\}$ is a function on $S \cup \{\alpha_0\}$; it lies in $\mathcal{F}$ and is strictly greater than f, contradicting maximality. So $S = A$, and $f \in \prod_{\alpha \in A} X_\alpha$.

Definition

ZFC := ZF + AoC, i.e. Zermelo–Fraenkel set theory with the Axiom of Choice as an axiom.

2.7 lim sup and lim inf of Sets

Notation. For $\{E_n\}_{n=1}^{\infty} \subseteq \mathcal{P}(X)$, set

$$\limsup_n E_n := \bigcap_{k=1}^{\infty} \bigcup_{n=k}^{\infty} E_n, \qquad \liminf_n E_n := \bigcup_{k=1}^{\infty} \bigcap_{n=k}^{\infty} E_n.$$

The notation comes from

$$\limsup_n x_n = \lim_{k\to\infty} \sup_{n\geq k} x_n, \qquad \liminf_n x_n = \lim_{k\to\infty} \inf_{n\geq k} x_n.$$

For sets, $\sup_{n\geq k} E_n := \bigcup_{n\geq k} E_n$ decreases as k increases, and $\inf_{n\geq k} E_n := \bigcap_{n\geq k} E_n$ increases.

Note. In terms of points,

$$\limsup_n E_n = \{x \mid x \in E_n \text{ for infinitely many } n\},$$
$$\liminf_n E_n = \{x \mid x \in E_n \text{ for all but finitely many } n\}.$$

Explanation. For example, if $E_n = A$ for even n and $E_n = B$ for odd n, then $\limsup_n E_n = A \cup B$ and $\liminf_n E_n = A \cap B$. In general $\liminf_n E_n \subseteq \limsup_n E_n$, since "all but finitely many n" implies "infinitely many n".

Definition

The same notation is used for a function at a point. Let $f : D \to \mathbb{R}$ with $D \subseteq \mathbb{R}$, and let x be a limit point of D. The *limit superior* and *limit inferior of f at x* are

$$\limsup_{y\to x} f(y) := \lim_{\delta\searrow 0} \sup\{f(y) \mid y \in D,\ 0 < |y - x| < \delta\},$$
$$\liminf_{y\to x} f(y) := \lim_{\delta\searrow 0} \inf\{f(y) \mid y \in D,\ 0 < |y - x| < \delta\},$$

with values in $[-\infty, \infty]$. The one-sided versions $\limsup_{y\to x^+} f(y)$ and $\limsup_{y\to x^-} f(y)$ use only $y > x$, resp. $y < x$, and likewise for $\liminf$.

Explanation. Both limits exist in $[-\infty, \infty]$: as δ shrinks, the supremum is taken over fewer points, so it decreases, and the infimum increases. So $\limsup_{y\to x} f(y)$ is the largest value that f keeps approaching near x, and $\liminf_{y\to x} f(y)$ the smallest; the value $f(x)$ itself plays no role. Always $\liminf_{y\to x} f(y) \leq \limsup_{y\to x} f(y)$, and $\lim_{y\to x} f(y)$ exists in $[-\infty, \infty]$ exactly when the two agree. For $f(y) = \sin(1/y)$ on $D = \mathbb{R} \setminus \{0\}$ and $x = 0$, $\limsup_{y\to 0} f(y) = 1$ and $\liminf_{y\to 0} f(y) = -1$. For an increasing $F : \mathbb{R} \to \mathbb{R}$ the one-sided limits exist: $F(x^-) := \lim_{y\to x^-} F(y) = \sup_{y<x} F(y)$ and $F(x^+) = \inf_{y>x} F(y)$, with $F(x^-) \leq F(x) \leq F(x^+)$. This notation is used for distribution functions in Section 8.

2.8 Sums over Arbitrary Index Sets

Definition

Let X be an index set (possibly uncountable) and $f : X \to [0, \infty]$. Define

$$\sum_{x\in X} f(x) := \sup\left\{ \sum_{x\in F} f(x) \;\middle|\; F \subseteq X,\ F \text{ finite} \right\}.$$

Explanation. For $X = \mathbb{N}$ this is the usual sum of a series of non-negative terms; the order of the terms does not matter since they are all ≥ 0. If the sum is finite, then $f(x) > 0$ for at most countably

many x, because for each n only finitely many x can have $f(x) > 1/n$. These sums are used for counting measures (Section 5).

2.9 The Measure Problem

Motivation. Suppose $E \subseteq \mathbb{R}^n$. Refining the Overview in Section 1, we want to define the measure $\mu(E)$ of E so that it agrees with the usual Euclidean volume when E is sufficiently regular, and has the following properties:

① $0 \le \mu(E)$.
② If $\{E_k\}_{k=1}^{\infty}$ are pairwise disjoint subsets of $\mathbb{R}^n$ ($E_k \cap E_\ell = \emptyset$ for $k \neq \ell$), then

$$\mu\Big(\bigcup_{k=1}^{\infty} E_k\Big) = \sum_{k=1}^{\infty} \mu(E_k).$$

③ $E \subseteq F \implies \mu(E) \le \mu(F)$.
④ If $E \cong F$ (E is congruent to F under a Euclidean isometry), then $\mu(E) = \mu(F)$.
⑤ For the unit cube $Q = \{x = (x_1, \dots, x_n) \in \mathbb{R}^n \mid 0 \le x_j \le 1,\ j = 1, \dots, n\}$, $\mu(Q) = 1$.

Question. Does such a measure μ exist?

Answer. No. We have to find the collection of "measurable" subsets $E \subseteq \mathbb{R}^n$: μ cannot be defined on *all* subsets.

3 Non-Measurable Sets and σ-Algebras

3.1 A Non-Measurable Set

The proof of the Answer uses the AoC. With it we construct a so-called *Vitali set* and show that no $\mu : \mathcal{P}(\mathbb{R}) \to [0, \infty]$, defined on *all* subsets, satisfies ②, ④ (the Lebesgue case) and ⑤. The other two properties are then automatic: ① is built into the target $[0, \infty]$, and taking $E_1 = Q$ and $E_k = \emptyset$ for $k \ge 2$ in ② gives $\mu(\emptyset) = 0$, so ② also holds for finite disjoint unions; in particular $\mu(F) = \mu(E) + \mu(F \setminus E) \ge \mu(E)$ for $E \subseteq F$, which is ③.

Counterexample (Vitali; $n = 1$)

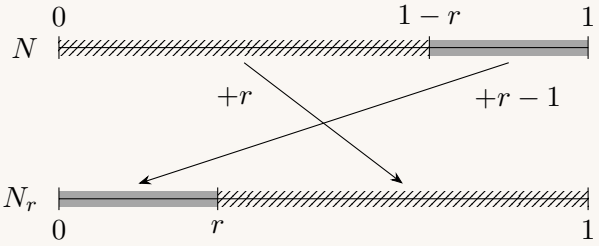
Let $I = [0, 1)$, and define an equivalence relation $\sim$ on I by

$$x \sim y \iff x - y \in \mathbb{Q} \qquad (x, y \in I).$$

Let $N \subseteq [0, 1)$ be a set containing precisely one element from each equivalence class (a *Vitali set*). (It exists by the AoC, applied to the family of distinct equivalence classes, which are nonempty and disjoint with union I.) Set $R = \mathbb{Q} \cap [0, 1)$, and for each $r \in R$ define

$$N_r := \{x + r \mid x \in N \cap [0, 1 - r)\} \cup \{x + r - 1 \mid x \in N \cap [1 - r, 1)\},$$

i.e. translate N by r and move the part that sticks out of $[0, 1)$ back by 1:



By construction, $N_r \subseteq [0,1)$ for all $r \in R$.

Each $x \in [0,1)$ belongs to exactly one N_r. Let $y \in N$ be the element with $y \sim x$. If $x \geq y$, then $r := x - y \in R$ and $y = x - r \in [0, 1-r)$, so $x = y + r \in N_r$. If $x < y$, then $r := x - y + 1 \in R$ and $y = x - r + 1 \in [1-r, 1)$, so $x = y + r - 1 \in N_r$. Conversely, suppose $x \in N_r \cap N_s$ with $r, s \in R$. Then $x - r$ or $x - r + 1$, and $x - s$ or $x - s + 1$, are elements of N in the class of x, so they are equal. Hence $r - s \in \{-1, 0, 1\}$, and since $|r - s| < 1$, $r = s$.

Now suppose such a μ exists. For $r \in R$, N is the disjoint union of $N \cap [0, 1-r)$ and $N \cap [1-r, 1)$, and N_r is the disjoint union of their translates by r and by $r - 1$. By additivity and translation invariance,

$$\mu(N) = \mu\big(N \cap [0, 1-r)\big) + \mu\big(N \cap [1-r, 1)\big) = \mu(N_r).$$

Since R is countable and $[0,1) = \bigcup_{r \in R} N_r$ is a disjoint union,

$$1 = \mu\big([0,1)\big) = \sum_{r \in R} \mu(N_r) = \sum_{r \in R} \mu(N). \qquad (*)$$

(Here $\mu([0,1)) = \mu([0,1]) - \mu(\{1\}) = 1$: $\mu([0,1]) = 1$ by ⑤; by ④ all one-point sets have the same measure, and since $[0,1]$ contains infinitely many of them, this measure is 0 by ② and ③.) There are only two possibilities, and both contradict $(*)$:

① $\mu(N) > 0$: the right side of $(*)$ is ∞;
② $\mu(N) = 0$: the right side of $(*)$ is $0 \neq 1$.

Hence no such μ exists.

Remark. We use $[0,1)$ throughout: with $[0,1]$, both $0, 1 \in R$ and N_0, N_1 both contain $N \cap [0,1)$, so "exactly one N_r" fails.

Explanation. The argument applies μ only to N, its pieces $N \cap [0, 1-r)$ and $N \cap [1-r, 1)$, their translates, and intervals. Lebesgue measure will be defined on a translation-invariant σ-algebra containing the intervals, and it will itself be translation invariant (Section 9). So if N were Lebesgue measurable, all these sets would be too, and the same contradiction would follow: N is the standard example of a non-measurable set.

Ansatz. (Rough.) Start with "elementary" sets, which form an *algebra*. Then close up under countable unions, countable intersections and complements, which gives a *σ-algebra*. By constructing an "appropriate" σ-algebra we get the measurable sets.

3.2 Algebras and σ-Algebras

Let $X \neq \emptyset$.

Definition

An *algebra* $\mathcal{A}$ on X is a nonempty collection of subsets of X such that

① $E_1, \dots, E_n \in \mathcal{A} \implies \bigcup_{j=1}^{n} E_j \in \mathcal{A}$ and $\bigcap_{j=1}^{n} E_j \in \mathcal{A}$;
② $E \in \mathcal{A} \implies E^c \in \mathcal{A}$.

Example

$X = \mathbb{R}^n$. An algebra will be constructed from cubes Q. This is the Riemann setting: for finite unions and intersections of cubes we know how to compute the volume. The new ingredient is the extension to σ-algebras.

Definition

An algebra $\mathcal{A}$ is a *σ-algebra* if it is closed under *countable* unions, intersections and complements, i.e. $E \in \mathcal{A} \implies E^c \in \mathcal{A}$, and

$$\{E_j\}_{j=1}^{\infty} \subseteq \mathcal{A} \implies \bigcup_{j=1}^{\infty} E_j \in \mathcal{A} \quad \text{and} \quad \bigcap_{j=1}^{\infty} E_j \in \mathcal{A}.$$

Note. For $\{E_j\}_{j=1}^{\infty} \subseteq \mathcal{A}$,

$$\bigcap_{j=1}^{\infty} E_j = \Big(\bigcup_{j=1}^{\infty} E_j^c \Big)^c,$$

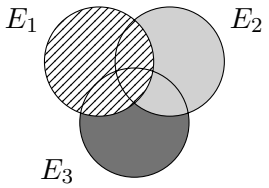
so closure under countable unions and complements implies closure under countable intersections, and likewise for finite ones. Hence in both definitions the conditions on intersections can be dropped.

Explanation. Every algebra contains $\emptyset$ and X: pick any $E \in \mathcal{A}$ (possible since $\mathcal{A} \neq \emptyset$); then $X = E \cup E^c \in \mathcal{A}$ and $\emptyset = X^c \in \mathcal{A}$. Not every algebra is a σ-algebra. On $\mathbb{R}$, the finite unions of intervals (points and unbounded intervals allowed) form an algebra, but $\mathbb{Q} = \bigcup_{q \in \mathbb{Q}} \{q\}$ is a countable union of points and not a finite union of intervals.

Claim (Disjoint unions)

Let $E_j \subseteq X$ for $j \in \mathbb{N}$, and set $F_k := E_k \setminus \bigcup_{j=1}^{k-1} E_j$. Then the F_k are pairwise disjoint and

$$\bigcup_{j=1}^{\infty} E_j = \bigcup_{k=1}^{\infty} F_k.$$



$F_1 = E_1$ (hatched)
$F_2 = E_2 \setminus E_1$ (light)
$F_3 = E_3 \setminus (E_1 \cup E_2)$ (dark)
and so on.

Proof

If $k > \ell$, then $F_k \subseteq E_\ell^c$ while $F_\ell \subseteq E_\ell$, so $F_k \cap F_\ell = \emptyset$. Each $F_k \subseteq E_k$; conversely, a point $x \in \bigcup_j E_j$ lies in F_k for the least k with $x \in E_k$.

In particular, if $\mathcal{A}$ is an algebra and all $E_j \in \mathcal{A}$, then $F_k = E_k \cap \left(\bigcup_{j<k} E_j\right)^c \in \mathcal{A}$ for every k. So it is enough to consider countable *disjoint* unions: an algebra is a σ-algebra as soon as it is closed under countable disjoint unions.

Example

Each of the following is a σ-algebra on X:

① $\mathcal{P}(X)$;
② $\{\emptyset, X\}$;
③ $\mathcal{A} = \{E \subseteq X \mid E \text{ is countable or } E^c \text{ is countable}\}$.

Only ③ needs a check. Complements swap the two conditions. A countable union of members of $\mathcal{A}$ is countable if all of them are; otherwise some member E has E^c countable, and the complement of the union lies in E^c.

3.3 Generated σ-Algebras

Claim (Basic fact)

If $\{\mathcal{A}_\alpha\}_{\alpha \in J}$ is a nonempty family of σ-algebras on X, then $\bigcap_{\alpha \in J} \mathcal{A}_\alpha$ is a σ-algebra on X.

Proof

Each $\mathcal{A}_\alpha$ contains X and is closed under complements and countable unions, so the intersection is too.

Definition

For $\mathcal{E} \subseteq \mathcal{P}(X)$, the *$\sigma$-algebra generated by $\mathcal{E}$* is the intersection of all σ-algebras containing $\mathcal{E}$:

$$\mathcal{M}(\mathcal{E}) := \bigcap \{\mathcal{A} \mid \mathcal{A} \text{ is a } \sigma\text{-algebra on } X \text{ and } \mathcal{E} \subseteq \mathcal{A}\}.$$

The family is nonempty since it contains $\mathcal{P}(X)$, so by the Basic fact $\mathcal{M}(\mathcal{E})$ is a σ-algebra, and it is the smallest one containing $\mathcal{E}$. These generated σ-algebras will be of particular interest to us.

Lemma

Let $\mathcal{E}, \mathcal{F} \subseteq \mathcal{P}(X)$ be families of subsets of X with $\mathcal{E} \subseteq \mathcal{M}(\mathcal{F})$. Then $\mathcal{M}(\mathcal{E}) \subseteq \mathcal{M}(\mathcal{F})$.

Proof

$\mathcal{M}(\mathcal{F})$ is a σ-algebra containing $\mathcal{E}$, and $\mathcal{M}(\mathcal{E})$ is the smallest such σ-algebra. Hence $\mathcal{M}(\mathcal{E}) \subseteq \mathcal{M}(\mathcal{F})$.

Explanation. $\mathcal{M}(\mathcal{E})$ is defined from above, as an intersection, and there is usually no explicit list of its members. What we use instead is the Lemma: to show $\mathcal{M}(\mathcal{E}) \subseteq \mathcal{A}$ for a σ-algebra $\mathcal{A}$, it is enough to check $\mathcal{E} \subseteq \mathcal{A}$. All the results on generators below (Borel sets, product σ-algebras) are proved this way.

3.4 Borel σ-Algebras

Let (X, d) be a metric space, $X \neq \emptyset$. There is then a distinguished σ-algebra that arises from the metric space structure.

Definition

The *Borel σ-algebra* $\mathcal{B}_X$ is the σ-algebra generated by the open subsets of X. Equivalently, it is generated by the closed subsets of X (by the Lemma, since closed sets are the complements of open sets).

Note. $\mathcal{B}_X$ contains the open sets, the closed sets, and countable unions and intersections of both. A countable union of open sets U_j is again open, and a countable intersection of closed sets F_k is again closed. However,

$$\bigcap_{j=1}^{\infty} U_j \in \mathcal{B}_X \quad (U_j \text{ open}), \qquad \bigcup_{k=1}^{\infty} F_k \in \mathcal{B}_X \quad (F_k \text{ closed}),$$

and these need not be open or closed. They are called *G_δ sets* and *F_σ sets*, respectively. E.g. $[a, b) = \bigcup_{n=1}^{\infty} [a, b - \frac{1}{n}]$ is an F_σ set that is neither open nor closed.

Borel sets are studied in much more depth in MATH 592 (descriptive set theory); see my notes for that course.

The main special case is the Lebesgue case:

$$X = \mathbb{R}^n, \qquad d(x, y) = \|x - y\| = \Big(\sum_{j=1}^{n} (x_j - y_j)^2 \Big)^{1/2}.$$

For $n = 1$, $X = \mathbb{R}$ and $d(x, y) = |x - y|$, and $\mathcal{B}_{\mathbb{R}}$ can be characterized as follows.

Proposition (Generators of $\mathcal{B}_{\mathbb{R}}$)

$\mathcal{B}_{\mathbb{R}}$ is generated by each of the following families:

(a) open intervals $\mathcal{E}_1 = \{(a, b) \mid a < b\}$;

ⓑ closed intervals $\mathcal{E}_2 = \{[a,b] \mid a < b\}$;
ⓒ half-open intervals $\mathcal{E}_3 = \{(a,b] \mid a < b\}$ or $\mathcal{E}_4 = \{[a,b) \mid a < b\}$;
ⓓ open rays $\mathcal{E}_5 = \{(a,\infty) \mid a \in \mathbb{R}\}$ or $\mathcal{E}_6 = \{(-\infty,a) \mid a \in \mathbb{R}\}$;
ⓔ closed rays $\mathcal{E}_7 = \{[a,\infty) \mid a \in \mathbb{R}\}$ or $\mathcal{E}_8 = \{(-\infty,a] \mid a \in \mathbb{R}\}$.

4 Borel and Product σ-Algebras

4.1 Generators of $\mathcal{B}_{\mathbb{R}}$

Proof

ⓐ We show $\mathcal{B}_{\mathbb{R}} = \mathcal{M}(\mathcal{E}_1)$. Each (a,b) is open, so $\mathcal{E}_1 \subseteq \mathcal{B}_{\mathbb{R}}$, and $\mathcal{M}(\mathcal{E}_1) \subseteq \mathcal{B}_{\mathbb{R}}$ by the Lemma.

Conversely, by the Lemma it is enough to show that every open set lies in $\mathcal{M}(\mathcal{E}_1)$. Key result (Folland, Chapter 0; the Corollary in Section 5): every open set in $\mathbb{R}$ is a countable disjoint union of open intervals. A bounded one is in $\mathcal{E}_1$, and an unbounded one is a countable union of bounded ones, e.g. $(a,\infty) = \bigcup_{n=1}^{\infty}(a, a+n)$. Hence every open set lies in $\mathcal{M}(\mathcal{E}_1)$, and $\mathcal{B}_{\mathbb{R}} \subseteq \mathcal{M}(\mathcal{E}_1)$.

The other parts are simple exercises (they were on HW 1). In each case, write the members of one family as countable unions or intersections of members of another, and use the Lemma. E.g.

$$(a,b] = \bigcap_{n=1}^{\infty}\left(a, b+\frac{1}{n}\right) \qquad (\text{a } G_\delta \text{ set}),$$

so $\mathcal{E}_3 \subseteq \mathcal{M}(\mathcal{E}_1)$ and hence $\mathcal{M}(\mathcal{E}_3) \subseteq \mathcal{B}_{\mathbb{R}}$. Also

$$(a,b) = \bigcup_{n > 2/(b-a)}\left[a+\frac{1}{n}, b-\frac{1}{n}\right] \in \mathcal{M}(\mathcal{E}_2),$$

so $\mathcal{E}_1 \subseteq \mathcal{M}(\mathcal{E}_2)$ and hence $\mathcal{B}_{\mathbb{R}} = \mathcal{M}(\mathcal{E}_1) \subseteq \mathcal{M}(\mathcal{E}_2)$.

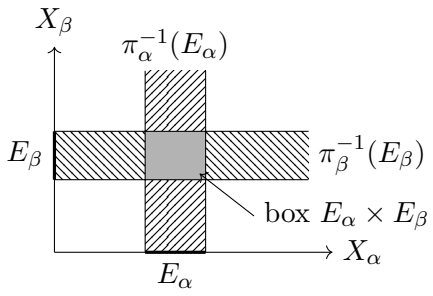
4.2 Product σ-Algebras

Let $\{X_\alpha\}_{\alpha\in A}$ be nonempty sets, $X = \prod_{\alpha\in A} X_\alpha$ their product set, and $\pi_\alpha : X \to X_\alpha$ the coordinate projections (Section 2).

Definition

Let $\mathcal{M}_\alpha$ be a σ-algebra on X_α for each $\alpha \in A$. The *product σ-algebra* $\bigotimes_{\alpha\in A}\mathcal{M}_\alpha$ on X is the σ-algebra generated by the *tube sets* $\pi_\alpha^{-1}(E_\alpha)$, i.e.

$$\bigotimes_{\alpha\in A}\mathcal{M}_\alpha := \mathcal{M}\big(\{\pi_\alpha^{-1}(E_\alpha) \mid E_\alpha \in \mathcal{M}_\alpha,\ \alpha \in A\}\big).$$



Explanation. A tube set restricts one coordinate only: $\pi_\alpha^{-1}(E_\alpha)$ consists of the points whose α-th coordinate lies in E_α, all other coordinates being free. For $X = \mathbb{R} \times \mathbb{R}$ the tubes are the strips $E \times \mathbb{R}$ and $\mathbb{R} \times F$. Once measurable maps are defined, the definition says that $\bigotimes_\alpha \mathcal{M}_\alpha$ is the smallest σ-algebra on X that makes every coordinate map π_α measurable.

The products $\prod_{\alpha \in A} E_\alpha$ with $E_\alpha \in \mathcal{M}_\alpha$ are the *box sets*. A box restricts all coordinates at once; in the figure, the box $E_\alpha \times E_\beta$ is where the two tubes meet. The key point is the following.

Proposition (Box sets)

Suppose A is *countable*. Then

$$\bigotimes_{\alpha \in A} \mathcal{M}_\alpha = \mathcal{M}\Big(\Big\{ \prod_{\alpha \in A} E_\alpha \;\Big|\; E_\alpha \in \mathcal{M}_\alpha \Big\} \Big).$$

Proof

We use the Lemma in both directions. Let $E_\alpha \in \mathcal{M}_\alpha$. Then

$$\pi_\alpha^{-1}(E_\alpha) = \prod_{\beta \in A} E_\beta, \qquad E_\beta := X_\beta \;\text{ for } \beta \neq \alpha,$$

is a box set, since $X_\beta \in \mathcal{M}_\beta$. So every tube set lies in the σ-algebra generated by the box sets, and by the Lemma $\bigotimes_\alpha \mathcal{M}_\alpha$ is contained in it.

Conversely, every box set is a countable intersection of tube sets (here we use that A is countable):

$$\prod_{\alpha \in A} E_\alpha = \bigcap_{\alpha \in A} \pi_\alpha^{-1}(E_\alpha) \in \bigotimes_{\alpha \in A} \mathcal{M}_\alpha.$$

By the Lemma, the σ-algebra generated by the box sets is contained in $\bigotimes_\alpha \mathcal{M}_\alpha$.

Explanation. For uncountable A the Prop can fail: a box with uncountably many factors $E_\alpha \neq X_\alpha$ is an intersection of uncountably many tubes, and it need not lie in $\bigotimes_\alpha \mathcal{M}_\alpha$. In fact, membership in a set of $\bigotimes_\alpha \mathcal{M}_\alpha$ always depends on only countably many coordinates.

We will need another proposition.

Proposition (Generators of $\bigotimes \mathcal{M}_\alpha$)

Suppose $\mathcal{M}_\alpha = \mathcal{M}(\mathcal{E}_\alpha)$, where $\mathcal{E}_\alpha$ is a collection of subsets of X_α, $\alpha \in A$. Then $\bigotimes_{\alpha\in A} \mathcal{M}_\alpha$ is generated by

$$\mathcal{F}_1 = \{\pi_\alpha^{-1}(E_\alpha) \mid E_\alpha \in \mathcal{E}_\alpha,\ \alpha \in A\}.$$

Moreover, if A is *countable* and $X_\alpha \in \mathcal{E}_\alpha$ for all $\alpha \in A$, then $\bigotimes_{\alpha\in A} \mathcal{M}_\alpha$ is generated by the box sets

$$\mathcal{F}_2 = \Big\{ \prod_{\alpha\in A} E_\alpha \;\Big|\; E_\alpha \in \mathcal{E}_\alpha \Big\}.$$

Proof

Since $\mathcal{E}_\alpha \subseteq \mathcal{M}_\alpha$, every member of $\mathcal{F}_1$ is a tube set, so $\mathcal{F}_1 \subseteq \bigotimes_\alpha \mathcal{M}_\alpha$ and $\mathcal{M}(\mathcal{F}_1) \subseteq \bigotimes_\alpha \mathcal{M}_\alpha$ by the Lemma.

For the other direction, $\bigotimes_\alpha \mathcal{M}_\alpha \subseteq \mathcal{M}(\mathcal{F}_1)$, consider for each $\alpha \in A$ the collection

$$\mathcal{N}_\alpha := \{E \subseteq X_\alpha \mid \pi_\alpha^{-1}(E) \in \mathcal{M}(\mathcal{F}_1)\}.$$

$\mathcal{N}_\alpha$ is a σ-algebra on X_α, since preimages under π_α commute with complements and countable unions (Section 2) and $\mathcal{M}(\mathcal{F}_1)$ is a σ-algebra.

Clearly $\mathcal{E}_\alpha \subseteq \mathcal{N}_\alpha$. Since $\mathcal{E}_\alpha$ generates $\mathcal{M}_\alpha$, the Lemma gives $\mathcal{M}_\alpha \subseteq \mathcal{N}_\alpha$, i.e. $\pi_\alpha^{-1}(E) \in \mathcal{M}(\mathcal{F}_1)$ for all $E \in \mathcal{M}_\alpha$. So every tube set lies in $\mathcal{M}(\mathcal{F}_1)$, and by the Lemma $\bigotimes_\alpha \mathcal{M}_\alpha \subseteq \mathcal{M}(\mathcal{F}_1)$.

The second statement follows as in the Prop (Box sets). Since $X_\beta \in \mathcal{E}_\beta$ for all β, each $\pi_\alpha^{-1}(E_\alpha)$ with $E_\alpha \in \mathcal{E}_\alpha$ is the box $\prod_\beta E_\beta \in \mathcal{F}_2$ with $E_\beta = X_\beta$ for $\beta \neq \alpha$; so $\mathcal{F}_1 \subseteq \mathcal{F}_2$. Conversely, each $\prod_\alpha E_\alpha \in \mathcal{F}_2$ equals $\bigcap_\alpha \pi_\alpha^{-1}(E_\alpha)$, a countable intersection of members of $\mathcal{F}_1$. By the Lemma, $\mathcal{M}(\mathcal{F}_2) = \mathcal{M}(\mathcal{F}_1) = \bigotimes_\alpha \mathcal{M}_\alpha$.

Remark. The lecture stated the second part without proof; the last paragraph of the proof above is added.

4.3 Product Borel σ-Algebras

Let (X_j, d_j), $j = 1, \ldots, n$, be a *finite* collection of metric spaces. The *product metric space* is $X = \prod_{j=1}^n X_j$ with the product metric

$$d(x, y) = \max_{j=1,\ldots,n} d_j(x_j, y_j), \qquad x = (x_1, \ldots, x_n),\ y = (y_1, \ldots, y_n).$$

Proposition

In this setting:

① $\bigotimes_{j=1}^n \mathcal{B}_{X_j} \subseteq \mathcal{B}_X$;
② $\bigotimes_{j=1}^n \mathcal{B}_{X_j} = \mathcal{B}_X$ if the X_j are *separable* (i.e. each has a countable dense subset).

Corollary

$\mathcal{B}_{\mathbb{R}^n} = \bigotimes_{j=1}^n \mathcal{B}_{\mathbb{R}} = \mathcal{B}_{\mathbb{R}} \otimes \cdots \otimes \mathcal{B}_{\mathbb{R}}$ (n times), where $\mathbb{R}^n$ has the Euclidean metric.

Proof

Apply ② with $X_j = \mathbb{R}$, which is separable ($\mathbb{Q}$ is dense). The max metric and the Euclidean metric on $\mathbb{R}^n$ have the same open sets, since $\max_j |x_j - y_j| \le \|x - y\| \le \sqrt{n}\, \max_j |x_j - y_j|$. So $\mathcal{B}_X = \mathcal{B}_{\mathbb{R}^n}$.

Note. Together with the explicit generators of $\mathcal{B}_{\mathbb{R}}$, the Corollary shows that $\mathcal{B}_{\mathbb{R}^n}$ is generated by the open boxes $\prod_{j=1}^n (a_j, b_j)$. Indeed, apply the Prop (Generators of $\bigotimes \mathcal{M}_\alpha$) with each $\mathcal{E}_j = \mathcal{E}_1 \cup \{\mathbb{R}\}$, which generates $\mathcal{B}_{\mathbb{R}}$ by ⓐ of the Prop (Generators of $\mathcal{B}_{\mathbb{R}}$) and contains $X_j = \mathbb{R}$. A box with some factors equal to $\mathbb{R}$ is a countable union of open boxes: replace each such factor by $(-k, k)$ and take the union over $k \in \mathbb{N}$.

The proof of ② below uses the following decomposition of open sets.

Lemma (Separable spaces)

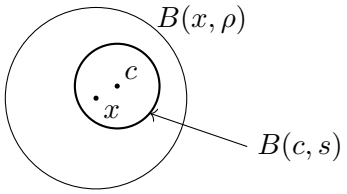
Let (Y, d) be a separable metric space, and let $C \subseteq Y$ be a countable dense subset. Then every open $U \subseteq Y$ is a countable union of balls:

$$U = \bigcup \big\{ B(c, s) \;\big|\; c \in C,\ s \in \mathbb{Q},\ s > 0,\ B(c, s) \subseteq U \big\}.$$

Proof

Let $x \in U$. Since U is open, there is $\rho > 0$ with $B(x, \rho) \subseteq U$. Choose a rational s with $0 < s < \rho/2$, and, by density, $c \in C$ with $d(x, c) < s$. Then

① $B(c, s) \subseteq B(x, 2s) \subseteq B(x, \rho) \subseteq U$, by the triangle inequality;
② $x \in B(c, s)$.



Since $x \in U$ is arbitrary, U is the union of all balls $B(c, s) \subseteq U$ with $c \in C$ and $s \in \mathbb{Q}$, $s > 0$, and there are only countably many such pairs (c, s).

Proof (of ①)

By the Prop (Generators of $\bigotimes \mathcal{M}_\alpha$), since the open sets generate each $\mathcal{B}_{X_j}$,

$$\bigotimes_{j=1}^{n} \mathcal{B}_{X_j} = \mathcal{M}\big(\{\pi_j^{-1}(U_j) \mid 1 \le j \le n,\ U_j \subseteq X_j \text{ open}\}\big).$$

Each $\pi_j : X \to X_j$ is continuous, since $d_j(x_j, y_j) \le d(x, y)$ by definition of the product metric. So the tube sets $\pi_j^{-1}(U_j)$ over open sets $U_j \subseteq X_j$ are open in X, and hence lie in $\mathcal{B}_X$. By the Lemma, $\bigotimes_j \mathcal{B}_{X_j} \subseteq \mathcal{B}_X$.

Proof (of ②)

By ① it remains to show $\mathcal{B}_X \subseteq \bigotimes_{j=1}^n \mathcal{B}_{X_j}$ when the X_j are separable. Let C_j be a countable dense subset of X_j, and let $\mathcal{E}_j$ be the collection of balls in X_j with centres in C_j and rational radii. By the Lemma (Separable spaces), every open set in X_j is a countable union of members of $\mathcal{E}_j$, so $\mathcal{B}_{X_j} = \mathcal{M}(\mathcal{E}_j)$. The set $C = \{x \in X \mid x_j \in C_j \ \forall j\}$ is countable and dense in X (given $x \in X$ and $\varepsilon > 0$, pick $c_j \in C_j$ with $d_j(x_j, c_j) < \varepsilon$; then $d(x, c) < \varepsilon$). So, by the same Lemma applied to X, every open set in X is a countable union of balls with centres in C and rational radii. For the max metric such a ball is a box,

$$\{y \in X \mid d(c, y) < r\} = \prod_{j=1}^n \{y_j \in X_j \mid d_j(c_j, y_j) < r\},$$

whose factors lie in $\mathcal{E}_j$. Hence $\mathcal{B}_X = \mathcal{M}\big(\{\prod_{j=1}^n E_j \mid E_j \in \mathcal{E}_j\}\big)$. Each such box is $\bigcap_{j=1}^n \pi_j^{-1}(E_j)$, a finite intersection of tube sets over $E_j \in \mathcal{E}_j \subseteq \mathcal{B}_{X_j}$, so it lies in $\bigotimes_j \mathcal{B}_{X_j}$. By the Lemma, $\mathcal{B}_X \subseteq \bigotimes_j \mathcal{B}_{X_j}$.

Explanation. Separability is what makes *countably* many boxes enough to build every open set, and it cannot simply be dropped. If $X_1 = X_2$ is a set with more points than $\mathbb{R}$, with the discrete metric ($d(x, y) = 1$ for $x \neq y$), then every subset of $X_1 \times X_2$ is open, but the diagonal $\{(x, x) \mid x \in X_1\}$ does not lie in $\mathcal{B}_{X_1} \otimes \mathcal{B}_{X_2}$.

Next. Abstract measures, and the construction of measures (Carathéodory construction, Sections 6–7).

5 Measures

5.1 Decomposition of Open Sets

Before making general definitions, we review a basic decomposition result. Open sets are of particular importance in measure theory, since they generate the Borel σ-algebra $\mathcal{B}_X$; in particular, they are central in the theory of Lebesgue measure. The key point is that open sets in separable metric spaces have a very nice decomposition into "elementary" sets, namely balls (the Lemma on separable spaces in Section 4). In $\mathbb{R}$ this can be sharpened as follows.

Corollary

Let $U \subseteq \mathbb{R}$ be open, where $\mathbb{R}$ has the usual metric $d(x, y) = |x - y|$. Then there is a countable collection $\{I_k\}$ of pairwise disjoint open intervals such that

$$U = \bigcup_k I_k.$$

Here "open interval" includes unbounded ones such as (a, ∞) and $\mathbb{R}$ itself. This is needed: $\mathbb{R}$ is open, but it is not a disjoint union of two or more nonempty open intervals.

Proof

The lecture left this as an exercise. For $x \in U$, let I_x be the union of all open intervals J with $x \in J \subseteq U$.

(1) *I_x is an open interval with $x \in I_x \subseteq U$.* It is nonempty, since U is open: $(x-r, x+r) \subseteq U$ for some $r > 0$. It is open, as a union of open sets. It is an interval: if $a < b$ lie in I_x, take intervals $J_a \ni a$ and $J_b \ni b$ from the union. Both contain x, so $J_a \cup J_b \supseteq [a, b]$.

(2) *Any two of these sets are equal or disjoint.* If $I_x \cap I_y \neq \emptyset$, then $I_x \cup I_y$ is an open interval contained in U and containing x, so it is one of the intervals J defining I_x. Hence $I_y \subseteq I_x$, and by symmetry $I_x = I_y$.

(3) *They cover U,* since $x \in I_x$ for every $x \in U$. So U is the disjoint union of the distinct sets I_x.

(4) *There are countably many.* Fix an enumeration $q_1, q_2, \ldots$ of $\mathbb{Q}$. Each I_x is a nonempty open interval, so by density of $\mathbb{Q}$ it contains some q_n; send it to the first such q_n. Distinct I_x's are disjoint, so they go to distinct rationals. This is an injection into $\mathbb{Q}$, and it uses no choice.

Labelling the distinct I_x's as $I_1, I_2, \ldots$ gives $U = \bigcup_k I_k$ with the I_k pairwise disjoint.

Explanation. Merging overlapping intervals two at a time never finishes when infinitely many overlap, e.g. $(1,3), (2,4), (3,5), \ldots$. The proof merges everything at once: I_x is the largest open interval around x inside U.

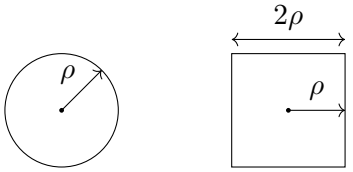
5.2 Open Sets in $\mathbb{R}^n$

On $\mathbb{R}^n$, $n > 1$, write $d_1(x, y) = \|x - y\|$ for the Euclidean metric of Section 3. Its balls are round: $B_{d_1}(0, \rho) = \{x \in \mathbb{R}^n \mid x_1^2 + \cdots + x_n^2 < \rho^2\}$.

Note. We can instead work with the product metric of Section 4,

$$d_2(x, y) := \max_{j=1,\ldots,n} |x_j - y_j|.$$

Then $B_{d_2}(0, \rho) = \{x \in \mathbb{R}^n \mid \max_j |x_j| < \rho\} = (-\rho, \rho)^n$ is an open cube of side 2ρ:



$B_{d_1}(0, \rho)$ $B_{d_2}(0, \rho)$

The metric d_2 has the same open sets as d_1 (see the proof of the Corollary in Section 4).

Similar to the 1-D case, cubes fit together rather nicely:

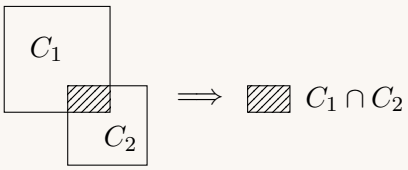
Definition

Two cubes C_1 and C_2 are *almost disjoint* if $\mathrm{Int}(C_1 \cap C_2) = \emptyset$, i.e. they can intersect only in boundary faces.

Explanation. E.g. $[0,1]^2$ and $[1,2] \times [0,1]$ share an edge, so they are almost disjoint but not disjoint. Faces have zero volume, so volumes still add up over almost disjoint unions; this is what makes the decomposition below useful.

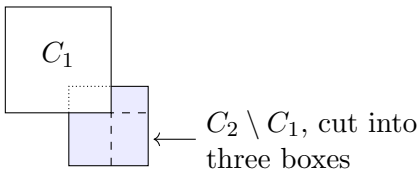
Example

($n = 2$) Two overlapping squares C_1 and C_2:



Their intersection is a rectangle with interior points, so C_1 and C_2 are not almost disjoint. In general two cubes meet in a box (a product of intervals), possibly empty, just as two intervals meet in an interval.

Note. $C_1 \cup C_2$ is the disjoint union of C_1 and $C_2 \setminus C_1$, and $C_2 \setminus C_1$ can be cut along the faces of C_1 into finitely many almost disjoint boxes (dashed lines):



$\Rightarrow$ any countable union of cubes (balls in d_2) can be written as an *almost* disjoint union of countably many cubes, which need not be open (this also follows from the Proposition below, as such a union is open).

This argument shows the following analogue of the 1-D result (the Corollary above).

Proposition

Let $U \subseteq \mathbb{R}^n$ be open. Then there is a countable collection of cubes $\{C_k\}$ such that

$$U = \bigcup_k C_k, \tag{$*$}$$

and the union in $(*)$ is almost disjoint. Here C_k is a cube but not necessarily open or closed:

$$C_k = \prod_{j=1}^{n} I_{j,k}, \qquad I_{j,k} \text{ intervals of equal length (open, closed or half-open).}$$

Proof

The lecture sketched this: by the Lemma on separable spaces in Section 4, applied with the metric d_2 and the dense subset $\mathbb{Q}^n$, U is a countable union of open cubes $B_{d_2}(c, s)$ with $c \in \mathbb{Q}^n$, $s \in \mathbb{Q}$, $s > 0$; the cutting argument above then makes them almost disjoint (left as an exercise). Cutting

produces boxes rather than cubes, so here is a complete proof that uses a fixed grid instead, in which two cubes never partly overlap.

For $k = 0, 1, 2, \ldots$, the *level-k grid cubes* are the closed cubes of side 2^{-k}

$$\prod_{j=1}^{n} \left[\frac{m_j}{2^k}, \frac{m_j + 1}{2^k}\right], \qquad m_1, \ldots, m_n \in \mathbb{Z}.$$

(1) *There are countably many grid cubes, and each level covers* $\mathbb{R}^n$: each level is indexed by $\mathbb{Z}^n$, and every real number lies in some $[m/2^k, (m+1)/2^k]$.

(2) *Two grid cubes are nested or almost disjoint.* Let $Q = \prod_j I_j$ have level k and $Q' = \prod_j J_j$ have level $l \geq k$. The endpoints of I_j are multiples of 2^{-k}, hence of 2^{-l}, while the interior of J_j contains no multiple of 2^{-l}. So either $J_j \subseteq I_j$, or I_j and J_j share at most an endpoint. If the second case occurs for some j, then $\operatorname{Int}(Q \cap Q') = \emptyset$; otherwise $Q' \subseteq Q$.

(3) *The grid cubes contained in U cover U.* Let $x \in U$. Since U is open, $B_{d_2}(x, r) \subseteq U$ for some $r > 0$. Pick k with $2^{-k} < r$. Every point y of a level-k cube containing x has $|y_j - x_j| \leq 2^{-k} < r$ for all j, so this cube lies in $B_{d_2}(x, r) \subseteq U$.

(4) *Each grid cube $Q \subseteq U$ lies in a maximal one*, i.e. one not contained in any larger grid cube inside U. Say Q has level k. A grid cube containing Q has level at most k, and there is at most one of each level: two of the same level that both contain Q overlap in the interior of Q, so by (2) they are nested, hence equal. So only finitely many grid cubes inside U contain Q, and the largest of them is maximal.

(5) *Distinct maximal cubes are almost disjoint*: they are not nested, since the smaller one would not be maximal, so (2) applies.

By (3) and (4), the maximal cubes cover U. So U is the union of the maximal grid cubes: countably many closed cubes, pairwise almost disjoint.

Explanation. The levels start at $k = 0$ on purpose: if grid cubes of every size were allowed, $U = \mathbb{R}^n$ would have no maximal cube. With half-open grid cubes $\prod_j [m_j/2^k, (m_j+1)/2^k)$ the same proof gives a disjoint union, since two such cubes are nested or disjoint. These cubes are neither open nor closed, which the statement allows.

This result is important since it allows us to define measures (premeasures) on open sets in $\mathbb{R}^n$, using countable additivity and the fact that we know the Euclidean volume of a cube: if C_k has side length ℓ_k, then $\mu(C_k) = \ell_k^n$, and

$$\mu(U) = \sum_k \mu(C_k).$$

5.3 Definition and Examples

Definition

Let $X \neq \emptyset$ and let $\mathcal{M}$ be a σ-algebra on X. A *measure* on $\mathcal{M}$ (or on $(X, \mathcal{M})$) is a function $\mu : \mathcal{M} \to [0, \infty]$ such that

① $\mu(\emptyset) = 0$;

② (Countable additivity) if $\{E_j\}_{j=1}^{\infty} \subseteq \mathcal{M}$ are pairwise disjoint, then

$$\mu\Big(\bigcup_{j=1}^{\infty} E_j\Big) = \sum_{j=1}^{\infty} \mu(E_j). \tag{†}$$

The triple $(X, \mathcal{M}, \mu)$ is then called a *measure space*.

Remark. μ is *finitely additive* if $\mu\big(\bigcup_{j=1}^{N} E_j\big) = \sum_{j=1}^{N} \mu(E_j)$ whenever $E_1, \dots, E_N \in \mathcal{M}$ are pairwise disjoint; e.g. area is finitely additive on elementary sets (finite unions of rectangles): the area of a union of disjoint elementary sets E_1, E_2, E_3 is the sum of their areas. Measures are finitely additive: take $E_j = \emptyset$ for $j > N$ in (†) and use $\mu(\emptyset) = 0$. However, there are finitely additive functions on $\mathcal{M}$ which are *not* countably additive.

Explanation. For example, on $\mathcal{M} = \mathcal{P}(\mathbb{N})$ let $\mu(E) = 0$ if E is finite and $\mu(E) = \infty$ if E is infinite. A union of finitely many sets is infinite iff one of them is, so μ is finitely additive. But $\mathbb{N} = \bigcup_n \{n\}$ gives $\mu(\mathbb{N}) = \infty \neq 0 = \sum_n \mu(\{n\})$, so μ is not a measure.

Example (1)

(Counting measures) Let $X \neq \emptyset$, $\mathcal{M} = \mathcal{P}(X)$ and $f : X \to [0, \infty]$. For $E \in \mathcal{P}(X)$ (i.e. $E \subseteq X$) set

$$\mu_f(E) := \sum_{x \in E} f(x), \tag{‡}$$

a sum over an arbitrary index set as in Section 2. Special cases of (‡):

① $f(x) = 1$ for all $x \in X$. Then $\mu_f(E) = \sum_{x \in E} 1$ is the number of elements of E (∞ if E is infinite). This is the *counting measure*.

② Given $x_0 \in X$, let $f(x) = 1$ if $x = x_0$ and $f(x) = 0$ if $x \neq x_0$. Then $\mu_f = \delta_{x_0}$, also written $\delta(x - x_0)$, is the *delta measure* at x_0: $\delta_{x_0}(E) = 1$ if $x_0 \in E$, and $\delta_{x_0}(E) = 0$ otherwise.

Claim

μ_f in (‡) is a measure on $\mathcal{P}(X)$.

Proof

Clearly $\mu_f(\emptyset) = 0$. Let $E_1, E_2, \dots$ be disjoint with union E. A finite $F \subseteq E$ splits into the finite sets $F \cap E_j$, so $\sum_{x \in F} f(x) \leq \sum_j \mu_f(E_j)$; taking the sup over F gives $\mu_f(E) \leq \sum_j \mu_f(E_j)$. Conversely, for finite $F_j \subseteq E_j$, $j \leq n$, the set $F_1 \cup \dots \cup F_n \subseteq E$ is finite, so $\sum_{j \leq n} \sum_{x \in F_j} f(x) \leq \mu_f(E)$; taking sups over the F_j and letting $n \to \infty$ gives $\sum_j \mu_f(E_j) \leq \mu_f(E)$.

5.4 Basic Properties

We will need the following basic properties of measures throughout the course.

Theorem

Let $(X, \mathcal{M}, \mu)$ be a measure space.

ⓐ (Monotonicity) If $E, F \in \mathcal{M}$ and $E \subseteq F$, then $\mu(E) \leq \mu(F)$.
ⓑ (Subadditivity) If $\{E_j\}_{j=1}^{\infty} \subseteq \mathcal{M}$, then $\mu\big(\bigcup_{j=1}^{\infty} E_j\big) \leq \sum_{j=1}^{\infty} \mu(E_j)$.
ⓒ (Continuity from below) If $\{E_j\}_{j=1}^{\infty} \subseteq \mathcal{M}$ and $E_1 \subseteq E_2 \subseteq E_3 \subseteq \cdots$, then

$$\mu\Big(\bigcup_{j=1}^{\infty} E_j\Big) = \lim_{j\to\infty} \mu(E_j).$$

ⓓ (Continuity from above) If $\{E_j\}_{j=1}^{\infty} \subseteq \mathcal{M}$, $E_1 \supseteq E_2 \supseteq E_3 \supseteq \cdots$ and $\mu(E_1) < \infty$, then

$$\mu\Big(\bigcap_{j=1}^{\infty} E_j\Big) = \lim_{j\to\infty} \mu(E_j).$$

Proof

ⓐ Let $E \subseteq F$. Then $F = E \cup (F \setminus E)$ is a disjoint union, so by (finite) additivity

$$\mu(F) = \mu(E) + \underbrace{\mu(F \setminus E)}_{\geq 0} \geq \mu(E).$$

ⓑ Write $\bigcup_{j=1}^{\infty} E_j$ as a countable *disjoint* union: by the Claim (Disjoint unions) in Section 3, the sets

$$F_1 = E_1, \qquad F_k = E_k \setminus \Big(\bigcup_{j=1}^{k-1} E_j\Big) \quad (k \geq 2)$$

lie in $\mathcal{M}$, are pairwise disjoint, and satisfy $\bigcup_k F_k = \bigcup_j E_j$. By additivity of μ,

$$\mu\Big(\bigcup_{j=1}^{\infty} E_j\Big) = \mu\Big(\bigcup_{k=1}^{\infty} F_k\Big) = \sum_{k=1}^{\infty} \mu(F_k). \tag{1}$$

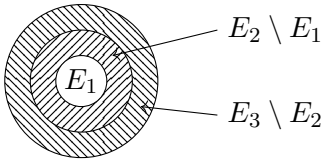
Since $F_k \subseteq E_k$, monotonicity gives

$$\mu(F_k) \leq \mu(E_k). \tag{2}$$

Substituting (2) into (1) gives $\mu\big(\bigcup_{j=1}^{\infty} E_j\big) \leq \sum_{j=1}^{\infty} \mu(E_j)$.

ⓒ We may assume $\mu(E_j) < \infty$ for all j (see the Note below). Set $E_0 := \emptyset$. Then

$$\bigcup_{j=1}^{\infty} E_j = \bigcup_{j=1}^{\infty} \underbrace{(E_j \setminus E_{j-1})}_{\text{disjoint}}.$$

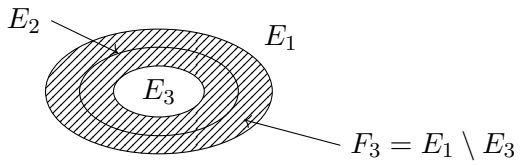


Since $E_j = E_{j-1} \cup (E_j \setminus E_{j-1})$ is a disjoint union and $\mu(E_{j-1}) < \infty$, additivity gives $\mu(E_j \setminus E_{j-1}) = \mu(E_j) - \mu(E_{j-1})$. Hence

$$\begin{aligned}
\mu\Big(\bigcup_{j=1}^{\infty} E_j\Big) &= \sum_{j=1}^{\infty} \mu(E_j \setminus E_{j-1}) = \lim_{n\to\infty} \sum_{j=1}^{n} \mu(E_j \setminus E_{j-1}) \\
&= \lim_{n\to\infty} \sum_{j=1}^{n} \big[\mu(E_j) - \mu(E_{j-1})\big] && \text{(telescoping sum)} \\
&= \lim_{n\to\infty} \mu(E_n) && \text{(since } \mu(E_0) = 0\text{)}.
\end{aligned}$$

ⓓ The idea is to take complements in E_1 and apply ⓒ. Set $F_j := E_1 \setminus E_j$ for $j = 1, 2, 3, \dots$, so $F_1 = \emptyset$, $F_2 = E_1 \setminus E_2$, $F_3 = E_1 \setminus E_3$, and

$$F_1 \subseteq F_2 \subseteq F_3 \subseteq \cdots, \quad \text{i.e. } F_j \nearrow \text{ as } j \to \infty.$$



Since $E_1 = F_j \cup E_j$ is a disjoint union, additivity gives

$$\mu(E_1) = \mu(F_j) + \mu(E_j). \tag{3}$$

Now $\bigcup_{j=1}^{\infty} F_j = E_1 \setminus \bigcap_{j=1}^{\infty} E_j$ (a point of E_1 lies in some F_j iff it misses some E_j), so again by additivity

$$\mu(E_1) = \mu\Big(\bigcup_{j=1}^{\infty} F_j\Big) + \mu\Big(\bigcap_{j=1}^{\infty} E_j\Big). \tag{4}$$

Since $F_j \nearrow$, ⓒ gives $\mu\big(\bigcup_j F_j\big) = \lim_j \mu(F_j)$. By (3), $\mu(F_j) = \mu(E_1) - \mu(E_j)$, because $\mu(E_j) \le \mu(E_1) < \infty$; and $\lim_j \mu(E_j)$ exists, since $\mu(E_j)$ decreases. So (4) becomes

$$\begin{aligned}
\mu(E_1) &= \lim_{j\to\infty} \big[\mu(E_1) - \mu(E_j)\big] + \mu\Big(\bigcap_{j=1}^{\infty} E_j\Big) \\
&= \mu(E_1) - \lim_{j\to\infty} \mu(E_j) + \mu\Big(\bigcap_{j=1}^{\infty} E_j\Big).
\end{aligned} \tag{5}$$

Since $\mu(E_1) < \infty$, we can subtract $\mu(E_1)$ from both sides of (5) and get ⓓ.

Note. In the proof of continuity from below it suffices to assume $\mu(E_j) < \infty$ for all j. Indeed, suppose $\mu(E_{j_0}) = \infty$ for some $j_0 \in \mathbb{N}$. Then by monotonicity $\mu\big(\bigcup_{j=1}^{\infty} E_j\big) \ge \mu(E_{j_0}) = \infty$, and also $\lim_{n\to\infty} \mu(E_n) = \infty$, since $E_n \supseteq E_{j_0}$ for all $n \ge j_0$. So both sides of ⓒ equal ∞.

Explanation. The hypothesis $\mu(E_1) < \infty$ in ⓓ cannot be dropped. For the counting measure on $\mathbb{N}$, the sets $E_j = \{j, j+1, j+2, \dots\}$ decrease and $\mu(E_j) = \infty$ for every j, but $\bigcap_j E_j = \emptyset$. It is enough that $\mu(E_k) < \infty$ for some k: apply ⓓ to $E_k \supseteq E_{k+1} \supseteq \cdots$, which has the same intersection and

the same limit.

6 Complete Measures and Outer Measures

6.1 Completeness of Measures

Definition

If $E \in \mathcal{M}$ and $\mu(E) = 0$, we say that E is a *null set* (of zero measure).

Remark. A statement holds *almost everywhere* (a.e.) if it holds on $X \setminus E$ for some null set E, i.e. for all x outside a set of measure zero.

Note. Suppose $E \in \mathcal{M}$ with $\mu(E) = 0$, and $F \subseteq E$ *with* $F \in \mathcal{M}$. Then by monotonicity $\mu(F) \leq \mu(E) = 0$, and since $\mu(F) \geq 0$, $\mu(F) = 0$.

Problem. Given $E \in \mathcal{M}$ with $\mu(E) = 0$, a subset $F \subseteq E$ need not satisfy $F \in \mathcal{M}$. So, a priori, it makes no sense to write $\mu(F)$.

Definition

$(\mathcal{M}, \mu)$ is *complete* if for every $E \in \mathcal{M}$ with $\mu(E) = 0$, every $F \subseteq E$ belongs to $\mathcal{M}$ (and then $\mu(F) = 0$, by the Note).

Explanation. Completeness is not automatic: Lebesgue measure on the Borel σ-algebra is not complete, because the Cantor set has measure zero but has more subsets than there are Borel sets, so some of its subsets are not Borel.

The next result says that one can, without loss of generality, assume that measures are complete.

Theorem

Let $(X, \mathcal{M}, \mu)$ be a measure space and $\mathcal{N} = \{N \in \mathcal{M} \mid \mu(N) = 0\}$. Set

$$\overline{\mathcal{M}} := \{E \cup F \mid E \in \mathcal{M},\ F \subseteq N \text{ for some } N \in \mathcal{N}\},$$

the *completion* of $\mathcal{M}$. Then $\overline{\mathcal{M}}$ is a σ-algebra, and there is a unique measure $\bar{\mu}$ on $\overline{\mathcal{M}}$ that extends μ, i.e. with $\bar{\mu}|_{\mathcal{M}} = \mu$. Moreover, $\bar{\mu}$ is complete.

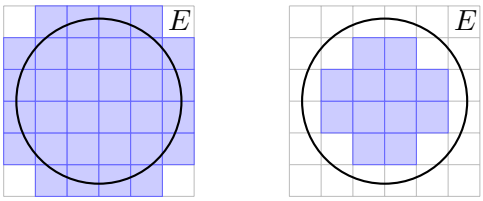
> ***Proof***
>
> Exercise (on HW 1; Folland, Exercise 1.6).

Remark. From now on, unless specified otherwise, we always assume that μ is complete. E.g. completing Lebesgue measure on the Borel sets gives Lebesgue measure on the Lebesgue measurable sets.

6.2 Outer and Inner Area

Ansatz. Start with a *premeasure* $\rho : \mathcal{A} \to [0, \infty]$, where $\mathcal{A}$ is an algebra of "elementary sets" (e.g. the algebra generated by cubes in $\mathbb{R}^n$). It has the properties of a measure, but is only defined on the algebra of elementary sets. To extend ρ beyond $\mathcal{A}$, we first define an *exterior* (outer) *measure* μ^* on *all* subsets $E \subseteq X$, using only the premeasure ρ on $\mathcal{A}$.

Heuristic. Cover $E \subseteq \mathbb{R}^2$ by the squares of a grid that meet E, and refine the grid: the total area of these squares tends to the "outer area" of E. Using only the squares contained in E instead gives the "inner area" of E.



outer area inner area

The outer measure $\mu^*(E)$ will generalize this simple notion of outer area from calculus. Similarly, one can define *inner measures* $\mu_*(E)$ that generalize the inner area.

Heuristic. A set E is "nicely behaved" (measurable) provided $\mu^*(E) = \mu_*(E)$.

Problem. This is the key problem: such a $\mu^* : \mathcal{P}(X) \to [0, \infty]$ need not be countably additive, i.e. it is not in general a measure.

The additional step will be to construct (Carathéodory) a particular σ-algebra $\mathcal{M} \subseteq \mathcal{P}(X)$ such that $\mu := \mu^*|_{\mathcal{M}}$ is a measure.

6.3 Outer Measures

Definition

A function $\mu^* : \mathcal{P}(X) \to [0, \infty]$ is an *outer measure* if

(1) $\mu^*(\emptyset) = 0$;
(2) $\mu^*(A) \leq \mu^*(B)$ if $A \subseteq B$;
(3) (Subadditivity) $\mu^*\big(\bigcup_{j=1}^{\infty} A_j\big) \leq \sum_{j=1}^{\infty} \mu^*(A_j)$ for all $\{A_j\}_{j=1}^{\infty} \subseteq \mathcal{P}(X)$.

Explanation. A simple outer measure that is not additive: if X has at least two points, set $\mu^*(\emptyset) = 0$ and $\mu^*(A) = 1$ for $A \neq \emptyset$. Properties (1)–(3) hold, but for $a \neq b$ we get $\mu^*(\{a, b\}) = 1 < 2 = \mu^*(\{a\}) + \mu^*(\{b\})$.

Proposition

Let $\mathcal{E} \subseteq \mathcal{P}(X)$ be a collection of "elementary sets" with $\emptyset, X \in \mathcal{E}$, and let $\rho : \mathcal{E} \to [0, \infty]$ satisfy

$\rho(\emptyset) = 0$ (e.g. a premeasure on an algebra $\mathcal{E}$, as in the Ansatz). For *any* $A \subseteq X$ set

$$\mu^*(A) := \inf\Big\{ \sum_{j=1}^{\infty} \rho(E_j) \;\Big|\; E_j \in \mathcal{E},\ A \subseteq \bigcup_{j=1}^{\infty} E_j \Big\}. \tag{†}$$

Then μ^* is an outer measure.

Remark. Compared with the outer area of the Heuristic, (†) allows *countable* coverings of A, which need not be disjoint.

Proof

μ^* is well-defined: for any $A \subseteq X$ there is a cover $\{E_j\}_{j=1}^{\infty} \subseteq \mathcal{E}$ of A (just take $E_j = X$ for all j), so (†) makes sense.

(1) Covering $\emptyset$ by $E_j = \emptyset$ for all j gives $0 \le \mu^*(\emptyset) \le \sum_j \rho(\emptyset) = 0$.

(2) Let $B \subseteq A$. If $\{E_j\}_{j=1}^{\infty} \subseteq \mathcal{E}$ covers A, then $B \subseteq A \subseteq \bigcup_j E_j$, i.e. covers of A are covers of B. So $\mu^*(B)$ is an infimum over at least as many sums as $\mu^*(A)$, and $\mu^*(B) \le \mu^*(A)$.

(3) (Subadditivity) Let $\{A_j\}_{j=1}^{\infty} \subseteq \mathcal{P}(X)$ and fix $\varepsilon > 0$. For each j, $\mu^*(A_j)$ is the infimum of $\sum_{k=1}^{\infty} \rho(E_j^k)$ over covers $\{E_j^k\}_{k=1}^{\infty} \subseteq \mathcal{E}$ of A_j, so there is such a cover with

$$\sum_{k=1}^{\infty} \rho(E_j^k) \le \mu^*(A_j) + \frac{\varepsilon}{2^j} \tag{1}$$

(if $\mu^*(A_j) = \infty$, any cover will do). Let $A = \bigcup_{j=1}^{\infty} A_j$. Since for each fixed j, $\{E_j^k\}_{k=1}^{\infty}$ covers A_j,

$$A \subseteq \bigcup_{j,k=1}^{\infty} E_j^k,$$

so $\{E_j^k\}_{j,k=1}^{\infty}$ is a countable cover of A (enumerate it as a sequence; a sum of non-negative terms does not depend on the order). Summing (1) over j,

$$\sum_{j,k} \rho(E_j^k) \le \sum_{j=1}^{\infty} \mu^*(A_j) + \underbrace{\sum_{j=1}^{\infty} \frac{\varepsilon}{2^j}}_{=\varepsilon}. \tag{2}$$

Since $\mu^*(A)$ is the infimum over all covers of A, (2) gives

$$\mu^*\Big(\bigcup_{j=1}^{\infty} A_j\Big) \le \sum_{j,k} \rho(E_j^k) \le \sum_{j=1}^{\infty} \mu^*(A_j) + \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary, letting $\varepsilon \to 0$ gives $\mu^*\big(\bigcup_{j=1}^{\infty} A_j\big) \le \sum_{j=1}^{\infty} \mu^*(A_j)$.

Explanation. Covering $E \in \mathcal{E}$ by $E, \emptyset, \emptyset, \ldots$ shows $\mu^*(E) \le \rho(E)$, but the inequality can be strict. On $X = \{1, 2\}$ with $\mathcal{E} = \{\emptyset, \{1\}, X\}$, $\rho(\{1\}) = 5$ and $\rho(X) = 1$, the cover $X, \emptyset, \emptyset, \ldots$ gives $\mu^*(\{1\}) \le 1$. When ρ is a premeasure on an algebra, $\mu^* = \rho$ on that algebra (Section 8).

6.4 μ^*-Measurable Sets

Definition

Let μ^* be an outer measure on X. A set $A \subseteq X$ is μ^*-*measurable* provided

$$\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c) \tag{$\dagger$}$$

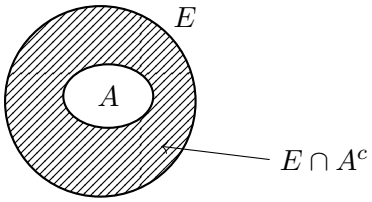
for every $E \subseteq X$.

Explanation. Since $E = (E\cap A)\cup(E\cap A^c)$, subadditivity always gives $\mu^*(E) \leq \mu^*(E\cap A)+\mu^*(E\cap A^c)$. So A is μ^*-measurable iff splitting any test set E along A never makes the outer measures of the two pieces add up to more than $\mu^*(E)$. For Lebesgue outer measure, the Vitali set of Section 3 fails this for some E.

Remark. Let $A \subseteq E$ with $\mu^*(E) < \infty$ (e.g. $\mu^*(X) < \infty$). Besides the *exterior measure* $\mu^*(A)$ of A (an outer approximation of A), consider the *interior measure* of A (relative to E),

$$\mu_*(A) := \mu^*(E) - \mu^*(E \cap A^c),$$

the outer measure of E minus that of the part of E outside A.



If A is μ^*-measurable, then, since $E \cap A = A$, ($\dagger$) gives

$$\mu^*(E) = \mu^*(A) + \mu^*(E \cap A^c) \iff \mu^*(A) = \mu_*(A),$$

as in the heuristic $\mu^* = \mu_*$.

Next. In Carathéodory's construction, the σ-algebra $\mathcal{M}$ will consist of the μ^*-measurable sets:

$$(\mathcal{E}, \rho) \longrightarrow (\mathcal{P}(X), \mu^*) \xrightarrow{\text{Carathéodory}} (\mathcal{M}, \mu^*|_{\mathcal{M}}).$$

7 The Carathéodory Construction

7.1 Carathéodory's Theorem

Theorem (Carathéodory)

Let μ^* be an outer measure on X, and let $\mathcal{M}$ be the collection of μ^*-measurable subsets of X. Then $\mathcal{M}$ is a σ-algebra, and $\mu := \mu^*|_{\mathcal{M}}$ is a complete measure on $\mathcal{M}$.

Proof

(i) $\mathcal{M}$ is an algebra. Clearly $\emptyset, X \in \mathcal{M}$, and $A \in \mathcal{M} \implies A^c \in \mathcal{M}$, since the definition of

μ^*-measurability is symmetric in A and A^c. Let $A, B \in \mathcal{M}$ and $E \subseteq X$. Using $A \in \mathcal{M}$, and then $B \in \mathcal{M}$ with the test sets $E \cap A$ and $E \cap A^c$,

$$\begin{aligned}\mu^*(E) &= \mu^*(E \cap A) + \mu^*(E \cap A^c) \\ &= \mu^*(E \cap A \cap B) + \mu^*(E \cap A \cap B^c) + \mu^*(E \cap A^c \cap B) + \mu^*(E \cap A^c \cap B^c). \end{aligned} \tag{1}$$

Since $A \cup B = (A \cap B) \cup (A \cap B^c) \cup (A^c \cap B)$, subadditivity gives

$$\mu^*\big(E \cap (A \cup B)\big) \leq \mu^*(E \cap A \cap B) + \mu^*(E \cap A \cap B^c) + \mu^*(E \cap A^c \cap B). \tag{2}$$

Putting (2) into (1) and using $A^c \cap B^c = (A \cup B)^c$,

$$\mu^*(E) \geq \mu^*\big(E \cap (A \cup B)\big) + \mu^*\big(E \cap (A \cup B)^c\big).$$

The reverse inequality is immediate from subadditivity. So $A \cup B \in \mathcal{M}$, and $\mathcal{M}$ is an algebra.

(ii) $\mu^*|_{\mathcal{M}}$ is finitely additive. Let $A, B \in \mathcal{M}$ with $A \cap B = \emptyset$. Testing $A \in \mathcal{M}$ with $E = A \cup B$,

$$\mu^*(A \cup B) = \mu^*\big((A \cup B) \cap A\big) + \mu^*\big((A \cup B) \cap A^c\big) = \mu^*(A) + \mu^*(B).$$

(iii) $\mathcal{M}$ is a σ-algebra and $\mu^*|_{\mathcal{M}}$ is countably additive. Since $\mathcal{M}$ is an algebra, it is enough to consider countable disjoint unions (Section 3). So let $\{A_j\}_{j=1}^{\infty} \subseteq \mathcal{M}$ be disjoint, and set

$$B_n := \bigcup_{j=1}^{n} A_j \quad (B_0 := \emptyset), \qquad B := \bigcup_{j=1}^{\infty} A_j,$$

and fix $E \subseteq X$. Since the A_j are disjoint, $B_n \cap A_n = A_n$ and $B_n \cap A_n^c = B_{n-1}$, so testing $A_n \in \mathcal{M}$ with $E \cap B_n$ gives

$$\mu^*(E \cap B_n) = \mu^*(E \cap A_n) + \mu^*(E \cap B_{n-1}),$$

and by induction

$$\mu^*(E \cap B_n) = \sum_{j=1}^{n} \mu^*(E \cap A_j). \tag{3}$$

Now $B_n \in \mathcal{M}$ by (i), and $B^c \subseteq B_n^c$, so by (3) and monotonicity

$$\mu^*(E) = \mu^*(E \cap B_n) + \mu^*(E \cap B_n^c) \geq \sum_{j=1}^{n} \mu^*(E \cap A_j) + \mu^*(E \cap B^c).$$

Let $n \to \infty$. Since $E \cap B = \bigcup_j (E \cap A_j)$, subadditivity then gives

$$\mu^*(E) \geq \sum_{j=1}^{\infty} \mu^*(E \cap A_j) + \mu^*(E \cap B^c) \geq \mu^*(E \cap B) + \mu^*(E \cap B^c) \geq \mu^*(E). \tag{4}$$

So equality holds throughout (4) for every $E \subseteq X$; in particular $B \in \mathcal{M}$. Taking $E = B$ in (4), where $B \cap A_j = A_j$ and $B \cap B^c = \emptyset$, gives $\mu^*(B) = \sum_{j=1}^{\infty} \mu^*(A_j)$. Hence $\mu^*|_{\mathcal{M}}$ is countably additive, and it is a measure since $\mu^*(\emptyset) = 0$.

(iv) μ is complete. Let $A \subseteq X$ with $\mu^*(A) = 0$. For any $E \subseteq X$, monotonicity gives $\mu^*(E \cap A) = 0$, so

$$\mu^*(E) \leq \mu^*(E \cap A) + \mu^*(E \cap A^c) = \mu^*(E \cap A^c) \leq \mu^*(E).$$

Hence $\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c)$, i.e. $A \in \mathcal{M}$: every set of outer measure zero is in $\mathcal{M}$. In particular, if $N \in \mathcal{M}$ with $\mu(N) = 0$ and $F \subseteq N$, then $\mu^*(F) = 0$, so $F \in \mathcal{M}$.

Explanation. The theorem holds for every outer measure, but $\mathcal{M}$ can be tiny. For the outer measure of Section 6 with $\mu^*(E) = 1$ for all $E \neq \emptyset$ (on a set X with at least two points), testing with $E = X$ shows that $\mathcal{M} = \{\emptyset, X\}$. Premeasures and metric outer measures (below) are what make $\mathcal{M}$ large.

7.2 Premeasures

Definition

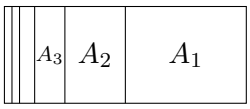
Let $\mathcal{A} \subseteq \mathcal{P}(X)$ be an algebra (of "elementary sets"). A *premeasure* on $\mathcal{A}$ is a function $\mu_0 : \mathcal{A} \to [0, \infty]$ such that

(1) $\mu_0(\emptyset) = 0$;
(2) if $\{A_j\}_{j=1}^{\infty} \subseteq \mathcal{A}$ are disjoint and $\bigcup_{j=1}^{\infty} A_j \in \mathcal{A}$, then

$$\mu_0\Big(\bigcup_{j=1}^{\infty} A_j\Big) = \sum_{j=1}^{\infty} \mu_0(A_j).$$

In particular, μ_0 is finitely additive (take $A_j = \emptyset$ for $j > n$ in (2)): for disjoint $A_1, \dots, A_n \in \mathcal{A}$,

$$\mu_0\Big(\bigcup_{j=1}^{n} A_j\Big) = \sum_{j=1}^{n} \mu_0(A_j).$$



$A = \bigcup_{j=1}^{\infty} A_j$

Explanation. In the picture, a half-open rectangle A is cut into countably many half-open strips A_j that pile up at its open left edge. For area on the algebra generated by half-open rectangles, A and all the A_j lie in $\mathcal{A}$, and (2) says that the areas of the strips add up to the area of A.

Here μ_0 plays the role of ρ (but also satisfies (2)), and $\mathcal{A}$ that of $\mathcal{E}$, in Section 6 (note $\emptyset, X \in \mathcal{A}$, since $\mathcal{A}$ is an algebra). So the construction of Section 6 turns μ_0 into an outer measure $\mu^* : \mathcal{P}(X) \to [0, \infty]$, the *induced outer measure* of μ_0.

7.3 Metric Outer Measures and Borel Regularity

Let (X, d) be a metric space and $\mathcal{B}_X$ its Borel σ-algebra. For $x \in X$ and $A, B \subseteq X$ write $d(x, B) := \inf_{b \in B} d(x, b)$ and $d(A, B) := \inf_{a \in A,\, b \in B} d(a, b)$.

Definition

Let μ^* be an outer measure on X.

① μ^* is a *metric outer measure* if $\mu^*(A \cup B) = \mu^*(A) + \mu^*(B)$ for all $A, B \subseteq X$ with $d(A, B) > 0$.
② μ^* is *Borel regular* if for every $E \subseteq X$ there is $B \in \mathcal{B}_X$ with $B \supseteq E$ and $\mu^*(B) = \mu^*(E)$.

Explanation. Additivity for *all* disjoint A, B would be too much: applied to $E \cap A$ and $E \cap A^c$ for every E, it would make every $A \subseteq X$ μ^*-measurable, which fails for Lebesgue outer measure.

Asking it only for sets a positive distance apart is the geometric compromise.

Proposition

Let μ^* be a metric outer measure on (X, d), with Carathéodory σ-algebra $\mathcal{M}$. Then $\mathcal{B}_X \subseteq \mathcal{M}$. Consequently $\overline{\mathcal{B}_X} \subseteq \mathcal{M}$, where $\overline{\mathcal{B}_X}$ is the completion of $\mathcal{B}_X$ w.r.t. the measure $\mu^*|_{\mathcal{B}_X}$.

Proof

Since $\mathcal{M}$ is a σ-algebra and $\mathcal{B}_X$ is generated by the closed sets, it is enough to show that every closed $F \subseteq X$ lies in $\mathcal{M}$. Then also $\overline{\mathcal{B}_X} \subseteq \mathcal{M}$, since $\mathcal{M}$ contains every set of outer measure zero (part (iv) of the proof of Carathéodory's theorem), in particular every subset of a Borel null set.

So let $F \subseteq X$ be closed and $A \subseteq X$. We need $\mu^*(A) = \mu^*(A \cap F) + \mu^*(A \setminus F)$. Here "$\leq$" is subadditivity, so it is enough to prove

$$\mu^*(A) \geq \mu^*(A \cap F) + \mu^*(A \setminus F). \tag{$\dagger$}$$

This is clear if $\mu^*(A) = \infty$, so assume $\mu^*(A) < \infty$. For $n \geq 1$ set

$$A_n := \left\{ x \in A \;\middle|\; d(x, F) \geq \frac{1}{n} \right\}, \qquad B_n := A_n \setminus A_{n-1} \quad (A_0 := \emptyset).$$

The A_n increase and lie in $A \setminus F$. Since F is closed, $d(x, F) > 0$ for every $x \notin F$ (otherwise $x \in \overline{F} = F$), so $A \setminus F = \bigcup_n A_n$. Hence the B_k are disjoint, and for every n

$$A \setminus F = \bigcup_{k=1}^{\infty} B_k = A_n \cup \bigcup_{k>n} B_k. \tag{1}$$

An outer measure need not be continuous along increasing sequences, so to get $\mu^*(A_n) \to \mu^*(A \setminus F)$ we use that μ^* is metric:

$$\sum_{k=1}^{\infty} \mu^*(B_k) < \infty \qquad \text{(Exercise)}.$$

(*Hint:* $d(B_j, B_k) > 0$ whenever $|j - k| \geq 2$; apply the metric property to the B_k with k even, and to those with k odd.)

Upshot: ① $\lim_{n\to\infty} \mu^*(A_n) = \mu^*(A \setminus F)$, and ② $d(A_n, A \cap F) \geq \frac{1}{n}$. For ①: by (1), monotonicity and subadditivity,

$$\mu^*(A_n) \leq \mu^*(A \setminus F) \leq \mu^*(A_n) + \sum_{k>n} \mu^*(B_k),$$

and the tail of the convergent series tends to 0. For ②: $d(x, y) \geq d(x, F) \geq \frac{1}{n}$ for $x \in A_n$ and $y \in A \cap F \subseteq F$.

By ② and the metric property, and then monotonicity ($A_n \cup (A \cap F) \subseteq A$),

$$\mu^*(A_n) + \mu^*(A \cap F) = \mu^*\big(A_n \cup (A \cap F)\big) \leq \mu^*(A).$$

Letting $n \to \infty$ and using ① gives ($\dagger$).

Theorem

Let μ^* be a metric outer measure on (X, d) that is Borel regular, and suppose $X = \bigcup_{k=1}^{\infty} X_k$ with $\mu^*(X_k) < \infty$ for all k. Then

$$\overline{\mathcal{B}_X} = \mathcal{M}.$$

Proof

"$\subseteq$" is the Prop. above. For "$\supseteq$", let $E \in \mathcal{M}$.

First suppose $\mu^*(E) < \infty$. By Borel regularity there is $B \in \mathcal{B}_X$ with $B \supseteq E$ and $\mu^*(B) = \mu^*(E)$. Testing $E \in \mathcal{M}$ with B,

$$\mu^*(B) = \mu^*(B \cap E) + \mu^*(B \setminus E) = \mu^*(E) + \mu^*(B \setminus E),$$

so $Z := B \setminus E$ has $\mu^*(Z) = 0$, as $\mu^*(E) < \infty$. By Borel regularity again, $Z \subseteq N$ for some $N \in \mathcal{B}_X$ with $\mu^*(N) = 0$. Since $B \setminus N \subseteq B \setminus Z = E$,

$$E = (B \setminus N) \cup (E \cap N),$$

where $B \setminus N \in \mathcal{B}_X$ and $E \cap N \subseteq N$. So $E \in \overline{\mathcal{B}_X}$.

In general, by Borel regularity we may take each $X_k \in \mathcal{B}_X$ (replace X_k by a Borel set $B_k \supseteq X_k$ with $\mu^*(B_k) = \mu^*(X_k)$). Then $E = \bigcup_k (E \cap X_k)$ with $E \cap X_k \in \mathcal{M}$ and $\mu^*(E \cap X_k) < \infty$. By the first case each $E \cap X_k \in \overline{\mathcal{B}_X}$, so $E \in \overline{\mathcal{B}_X}$, since $\overline{\mathcal{B}_X}$ is a σ-algebra.

Remark. The lecture stated the theorem without the assumption on the X_k, but some such assumption is needed: counting measure on $\mathbb{R}$ is a metric, Borel regular outer measure with $\mathcal{M} = \mathcal{P}(\mathbb{R})$, but its only null set is $\emptyset$, so $\overline{\mathcal{B}_{\mathbb{R}}} = \mathcal{B}_{\mathbb{R}} \neq \mathcal{M}$.

7.4 Lebesgue Measure on $\mathbb{R}^n$

On $\mathbb{R}^n$ take the max metric $d(x, y) := \max_{1 \le j \le n} |x_j - y_j|$ of Section 5. Let $m : \mathcal{A} \to [0, \infty]$ be the Lebesgue premeasure on the algebra $\mathcal{A}$ generated by the cubes Q, with $m(Q) = |Q|$ (Euclidean volume), and let $m^* : \mathcal{P}(\mathbb{R}^n) \to [0, \infty]$ be the induced outer measure, the *Lebesgue outer measure*. Since $\mathcal{A}$ is generated by cubes, m^* can be computed with covers by cubes: for $E \subseteq \mathbb{R}^n$,

$$m^*(E) = \inf \Big\{ \sum_{j=1}^{\infty} |Q_j| \;\Big|\; Q_j \text{ cubes}, \; E \subseteq \bigcup_{j=1}^{\infty} Q_j \Big\}. \tag{$*$}$$

Exercise

Show that m^* is a metric outer measure on $(\mathbb{R}^n, d)$. (*Hint:* cut each cube of a cover of $A \cup B$ into subcubes of side less than $d(A, B)$; none of them meets both A and B.)

Claim

m^* is Borel regular: every $E \subseteq \mathbb{R}^n$ is contained in a G_δ set B with $m^*(B) = m^*(E)$.

Proof

If $m^*(E) = \infty$, take $B = \mathbb{R}^n$. Otherwise fix $k \geq 1$. By $(*)$ there are cubes Q_j with $E \subseteq \bigcup_j Q_j$ and $\sum_j |Q_j| \leq m^*(E) + \frac{1}{2k}$. Enlarge each Q_j to an open cube $Q_j' \supseteq Q_j$ (same centre, slightly longer side) with $|Q_j'| \leq |Q_j| + \frac{2^{-j}}{2k}$. Then $U_k := \bigcup_j Q_j'$ is open, $U_k \supseteq E$, and by subadditivity and $m^*(Q_j') \leq |Q_j'|$ (cover Q_j' by itself in $(*)$),

$$m^*(U_k) \leq \sum_{j=1}^{\infty} m^*(Q_j') \leq \sum_{j=1}^{\infty} |Q_j'| \leq m^*(E) + \frac{1}{k}.$$

Set $B := \bigcap_{k=1}^{\infty} U_k$, a G_δ set, so $B \in \mathcal{B}_{\mathbb{R}^n}$. Then $B \supseteq E$ and $m^*(E) \leq m^*(B) \leq m^*(U_k) \leq m^*(E) + \frac{1}{k}$ for every k, so $m^*(B) = m^*(E)$.

Remark. In lecture the cubes were made almost disjoint and $\bigcup_j Q_j$ itself was taken as the open set; since a union of cubes need not be open, they are enlarged to open cubes above, and then subadditivity suffices.

Upshot. By the Exercise and the Claim, m^* is a metric, Borel regular outer measure on $(\mathbb{R}^n, d)$, and $\mathbb{R}^n = \bigcup_k (-k, k)^n$ with $m^*\big((-k,k)^n\big) \leq (2k)^n < \infty$. So by the theorem, the Carathéodory σ-algebra of m^* is

$$\mathcal{M} = \overline{\mathcal{B}_{\mathbb{R}^n}} =: \mathcal{L}_{\mathbb{R}^n},$$

the *Lebesgue σ-algebra*. The measure $m^*|_{\mathcal{L}_{\mathbb{R}^n}}$, again written m, is *Lebesgue measure*.

8 The Carathéodory Extension Theorem and Lebesgue–Stieltjes Measures

8.1 The Carathéodory Extension Theorem

Proposition (Carathéodory extension)

Let μ_0 be a premeasure on an algebra $\mathcal{A} \subseteq \mathcal{P}(X)$, with induced outer measure μ^* and Carathéodory σ-algebra $\mathcal{M}$. Then:

① $\mu^*|_{\mathcal{A}} = \mu_0$;
② every set in $\mathcal{A}$ is μ^*-measurable, i.e. $\mathcal{A} \subseteq \mathcal{M}$.

Proof

① Let $E \in \mathcal{A}$, and let $\{A_j\}_{j=1}^{\infty} \subseteq \mathcal{A}$ with $E \subseteq \bigcup_j A_j$. Set

$$B_n := E \cap \Big(A_n \setminus \bigcup_{j=1}^{n-1} A_j\Big), \qquad n = 1, 2, \ldots$$

By the Claim (Disjoint unions) in Section 3, the B_n are disjoint and $\bigcup_{n=1}^{\infty} B_n = E \cap \bigcup_n A_n = E$. Moreover $B_n \in \mathcal{A}$ (as $\mathcal{A}$ is an algebra) and $B_n \subseteq A_n$. Also, μ_0 is monotone on $\mathcal{A}$, by finite

additivity (as for measures, Section 5). So by (2) in the definition of a premeasure,

$$\mu_0(E) = \sum_{n=1}^{\infty} \mu_0(B_n) \le \sum_{n=1}^{\infty} \mu_0(A_n).$$

This holds for every cover of E by sets in $\mathcal{A}$, so taking the infimum gives $\mu_0(E) \le \mu^*(E)$. Conversely, $A_1 = E$, $A_j = \emptyset$ $(j \ge 2)$ is such a cover, so $\mu^*(E) \le \sum_j \mu_0(A_j) = \mu_0(E)$. Hence $\mu^*(E) = \mu_0(E)$.

② Let $A \in \mathcal{A}$, $E \subseteq X$ and $\varepsilon > 0$. By the definition of μ^* there are $B_j \in \mathcal{A}$ with $E \subseteq \bigcup_j B_j$ and $\sum_j \mu_0(B_j) \le \mu^*(E) + \varepsilon$. Since $B_j \cap A$ and $B_j \cap A^c$ are disjoint sets in $\mathcal{A}$ with union B_j, finite additivity of μ_0 gives

$$\begin{aligned} \mu^*(E) + \varepsilon \ge \sum_{j=1}^{\infty} \mu_0(B_j) &= \sum_{j=1}^{\infty} \mu_0(B_j \cap A) + \sum_{j=1}^{\infty} \mu_0(B_j \cap A^c) \\ &\ge \mu^*(E \cap A) + \mu^*(E \cap A^c), \end{aligned}$$

where the last step is the definition of μ^*: $\{B_j \cap A\}_j \subseteq \mathcal{A}$ covers $E \cap A$, and $\{B_j \cap A^c\}_j \subseteq \mathcal{A}$ covers $E \cap A^c$. Since $\varepsilon > 0$ is arbitrary, $\mu^*(E) \ge \mu^*(E \cap A) + \mu^*(E \cap A^c)$ for all $E \subseteq X$. The reverse inequality is subadditivity, so $A \in \mathcal{M}$.

Remark. So the measure $\mu^*|_{\mathcal{M}}$ of Carathéodory's theorem extends the premeasure μ_0; hence the name *Carathéodory extension.*

8.2 Uniqueness of the Extension

Definition

A measure space $(X, \mathcal{M}, \mu)$ is *σ-finite* if there are $E_j \in \mathcal{M}$ with $\bigcup_{j=1}^{\infty} E_j = X$ and $\mu(E_j) < \infty$ for all j. Likewise, a premeasure μ_0 on $\mathcal{A}$ is *σ-finite* if $X = \bigcup_{j=1}^{\infty} A_j$ with $A_j \in \mathcal{A}$ and $\mu_0(A_j) < \infty$ for all j.

Theorem (Carathéodory II)

Let μ_0 be a premeasure on an algebra $\mathcal{A} \subseteq \mathcal{P}(X)$, with induced outer measure μ^*, and let $\mathcal{M}(\mathcal{A})$ be the σ-algebra generated by $\mathcal{A}$. Then $\mu := \mu^*|_{\mathcal{M}(\mathcal{A})}$ is a measure on $\mathcal{M}(\mathcal{A})$ with $\mu|_{\mathcal{A}} = \mu_0$. Moreover, if μ_0 is σ-finite, then μ is the *unique* measure on $\mathcal{M}(\mathcal{A})$ that extends μ_0.

Proof

Existence. By the Carathéodory extension Prop., $\mathcal{A} \subseteq \mathcal{M}$; as $\mathcal{M}$ is a σ-algebra, $\mathcal{M}(\mathcal{A}) \subseteq \mathcal{M}$. So μ is the restriction of the measure $\mu^*|_{\mathcal{M}}$ (Carathéodory's theorem) to the smaller σ-algebra $\mathcal{M}(\mathcal{A})$, hence a measure, and $\mu|_{\mathcal{A}} = \mu_0$ by the same Prop.

Uniqueness. Let ν be another measure on $\mathcal{M}(\mathcal{A})$ with $\nu|_{\mathcal{A}} = \mu_0$, and let $E \in \mathcal{M}(\mathcal{A})$. If $E \subseteq \bigcup_j A_j$ with $A_j \in \mathcal{A}$, then by monotonicity and subadditivity of ν,

$$\nu(E) \le \sum_{j=1}^{\infty} \nu(A_j) = \sum_{j=1}^{\infty} \mu_0(A_j).$$

Taking the infimum over all such covers,

$$\nu(E) \le \mu^*(E) = \mu(E) \qquad \text{for all } E \in \mathcal{M}(\mathcal{A}). \tag{1}$$

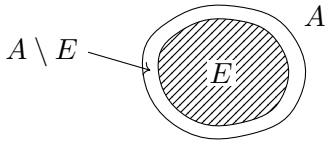
For the reverse inequality we argue in two steps.

Step 1: $\mu(E) < \infty$. Let $\varepsilon > 0$. Since $\mu(E) = \mu^*(E)$ is a finite infimum over covers of E by sets in $\mathcal{A}$, there are $A_j \in \mathcal{A}$ with $E \subseteq \bigcup_j A_j$ and $\sum_j \mu_0(A_j) < \mu(E) + \varepsilon$. Then $A := \bigcup_j A_j$ lies in $\mathcal{M}(\mathcal{A})$ and contains E, and by subadditivity

$$\mu(A) \le \sum_{j=1}^{\infty} \mu(A_j) = \sum_{j=1}^{\infty} \mu_0(A_j) < \mu(E) + \varepsilon. \tag{2}$$

Since $E \subseteq A$, additivity gives $\mu(A) = \mu(E) + \mu(A \setminus E)$. As $\mu(E) < \infty$, we can subtract:

$$\mu(A \setminus E) = \mu(A) - \mu(E) < \varepsilon \qquad \text{by (2).} \tag{3}$$



Moreover $\mu(A) = \nu(A)$: the sets $C_n := \bigcup_{j=1}^{n} A_j$ lie in $\mathcal{A}$ and increase to A, and $\mu(C_n) = \mu_0(C_n) = \nu(C_n)$, so by continuity from below (for μ and for ν)

$$\mu(A) = \lim_{n\to\infty} \mu(C_n) = \lim_{n\to\infty} \nu(C_n) = \nu(A).$$

Hence, using additivity of ν, then (1) for $A \setminus E \in \mathcal{M}(\mathcal{A})$, then (3),

$$\mu(E) \le \mu(A) = \nu(A) = \nu(E) + \nu(A \setminus E) \le \nu(E) + \mu(A \setminus E) < \nu(E) + \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary, $\mu(E) \le \nu(E)$; with (1), $\mu(E) = \nu(E)$.

Step 2: general E. Since μ_0 is σ-finite, $X = \bigcup_j A_j$ with $A_j \in \mathcal{A}$ and $\mu_0(A_j) < \infty$. Replacing A_j by $A_j \setminus \bigcup_{i<j} A_i \in \mathcal{A}$ (Section 3; still of finite premeasure), we may take the A_j disjoint. By monotonicity, $\mu(E \cap A_j) \le \mu(A_j) = \mu_0(A_j) < \infty$, so by countable additivity and Step 1,

$$\mu(E) = \sum_{j=1}^{\infty} \mu(E \cap A_j) = \sum_{j=1}^{\infty} \nu(E \cap A_j) = \nu(E).$$

Explanation. Without σ-finiteness, uniqueness can fail. On $X = \mathbb{Q}$, let $\mathcal{A}$ be the algebra generated by the sets $(a, b] \cap \mathbb{Q}$, whose nonempty members are all infinite, and let $\mu_0(\emptyset) = 0$ and $\mu_0(A) = \infty$ for $A \neq \emptyset$; this is a premeasure, and it is not σ-finite. Since each $\{q\} = \bigcap_n (q - \frac{1}{n}, q] \cap \mathbb{Q}$ lies in $\mathcal{M}(\mathcal{A})$ and $\mathbb{Q}$ is countable, $\mathcal{M}(\mathcal{A}) = \mathcal{P}(\mathbb{Q})$. Counting measure and twice counting measure both extend μ_0, yet they differ on singletons.

Note. The inclusion $\mathcal{M}(\mathcal{A}) \subseteq \mathcal{M}$ in the proof can be strict: the Carathéodory σ-algebra need not agree with the σ-algebra generated by $\mathcal{A}$.

8.3 Distribution Functions

Let $\mu : \mathcal{B}_{\mathbb{R}} \to [0, \infty]$ be a *finite* Borel measure on $\mathbb{R}$, i.e. $\mu(\mathbb{R}) < \infty$, and set

$$F(x) := \mu\big((-\infty, x]\big), \qquad x \in \mathbb{R}.$$

Then $F : \mathbb{R} \to \mathbb{R}$ (the values are finite since μ is finite); F is the *distribution function* of μ.

Claim

The function F is

① increasing;
② right continuous.

Proof

① If $y \geq x$, then $(-\infty, y] \supseteq (-\infty, x]$, so $F(y) \geq F(x)$ by monotonicity of μ.

② Let $x_n \searrow x$ (since F is increasing, such sequences suffice). Then the sets $(-\infty, x_n]$ decrease and $\bigcap_n (-\infty, x_n] = (-\infty, x]$. Since $\mu\big((-\infty, x_1]\big) < \infty$, continuity from above gives

$$F(x) = \mu\big((-\infty, x]\big) = \lim_{n\to\infty} \mu\big((-\infty, x_n]\big) = \lim_{n\to\infty} F(x_n).$$

Note. If $a < b$, then $(-\infty, b] = (-\infty, a] \cup (a, b]$ (disjoint), so

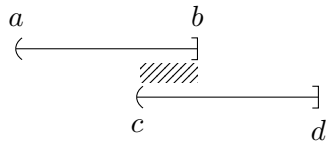
$$\mu\big((a, b]\big) = F(b) - F(a).$$

8.4 h-Intervals

Definition

An *h-interval* is either $\emptyset$ or a set $(a, b] \cap \mathbb{R}$ with $-\infty \leq a < b \leq \infty$, written $(a, b]$ for short. (So (a, ∞), $(-\infty, b]$ and $\mathbb{R}$ are h-intervals.)

Note. The intersection of two h-intervals is an h-interval, $(a, b] \cap (c, d] = (\max(a, c), \min(b, d)]$ (empty if $\max(a, c) \geq \min(b, d)$), and the complement of an h-interval is a disjoint union of at most two h-intervals, $(a, b]^c = (-\infty, a] \cup (b, \infty)$.



$$(a, b] \cap (c, d] = (c, b]$$

Set

$$\mathcal{A} := \{\text{finite disjoint unions of h-intervals}\}.$$

Note. $\mathcal{A}$ is an algebra. By the previous Note, the intersection of two sets in $\mathcal{A}$ is in $\mathcal{A}$ (intersect their h-intervals pairwise), and so is the complement of a set in $\mathcal{A}$ (intersect the complements of its h-intervals). Finally $A \cup B = (A^c \cap B^c)^c$.

Proposition

Let $F : \mathbb{R} \to \mathbb{R}$ be increasing and right continuous, and put $F(\pm\infty) := \lim_{x\to\pm\infty} F(x)$. For disjoint nonempty h-intervals $(a_j, b_j]$, $j = 1, \dots, n$, set

$$\mu_0\Big(\bigcup_{j=1}^{n}(a_j, b_j]\Big) := \sum_{j=1}^{n}\big[F(b_j) - F(a_j)\big], \qquad \mu_0(\emptyset) := 0.$$

Then μ_0 is a well-defined premeasure on $\mathcal{A}$.

Proof

The lecture left this as an exercise (Folland, Proposition 1.15). For a single h-interval, $\mu_0((a, b]) = F(b) - F(a)$.

(i) μ_0 is well defined and finitely additive. If an h-interval $(a, b]$ is the disjoint union of finitely many nonempty h-intervals, these line up end to end: ordered by left endpoints, they are $(a_j, b_j]$, $j = 1, \dots, n$, with $a = a_1 < b_1 = a_2 < b_2 = \cdots = a_n < b_n = b$. So the sum telescopes:

$$\sum_{j=1}^{n}\big[F(b_j) - F(a_j)\big] = F(b) - F(a). \tag{$*$}$$

Now let a set in $\mathcal{A}$ have two representations $\bigcup_i I_i = \bigcup_k J_k$ as finite disjoint unions of h-intervals. The sets $I_i \cap J_k$ are h-intervals; each I_i is the disjoint union of the $I_i \cap J_k$ over k, and each J_k is the disjoint union of the $I_i \cap J_k$ over i. Applying $(*)$ to each I_i and to each J_k,

$$\sum_i \mu_0(I_i) = \sum_{i,k} \mu_0(I_i \cap J_k) = \sum_k \mu_0(J_k).$$

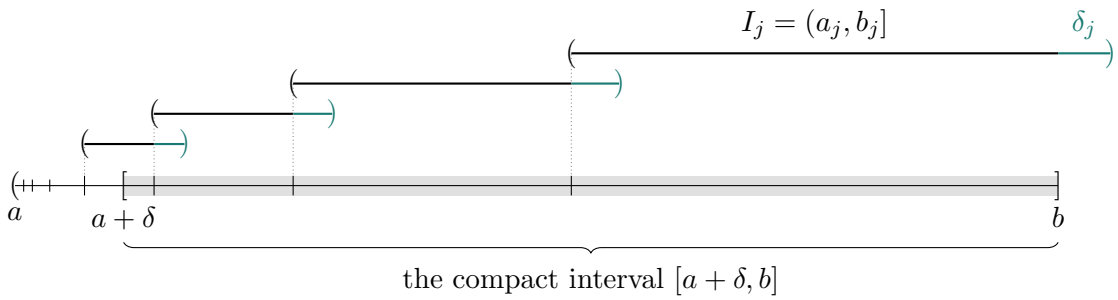
So μ_0 is well defined. It is finitely additive, since a finite disjoint union of sets in $\mathcal{A}$ is represented by putting their h-intervals together.

(ii) Reduction to one h-interval. Let $A_1, A_2, \ldots \in \mathcal{A}$ be disjoint with $A := \bigcup_j A_j \in \mathcal{A}$. Write A as a disjoint union of h-intervals $J_1, \dots, J_m$, and split each A_j into its h-intervals. Intersecting these with each J_k and using finite additivity (sums of non-negative terms can be rearranged), it is enough to prove: if an h-interval I is the disjoint union of h-intervals $I_j = (a_j, b_j]$, $j \in \mathbb{N}$, then

$$\mu_0(I) = \sum_{j=1}^{\infty} \mu_0(I_j). \tag{$\dagger$}$$

(iii) "$\geq$" in $(\dagger)$. For each n, $I \setminus (I_1 \cup \cdots \cup I_n) \in \mathcal{A}$, so by finite additivity $\mu_0(I) \geq \sum_{j=1}^{n} \mu_0(I_j)$. Let $n \to \infty$.

(iv) "$\leq$" in $(\dagger)$. This is where right continuity is used. The idea is compactness, but $(a, b]$ is not compact and the I_j are not open. So we shrink I a little on the left and lengthen each I_j a little on the right; by right continuity, this costs at most ε.



In the figure, the I_j pile up at a, so no finitely many of them cover I. After shrinking I to $[a+\delta, b]$ and lengthening each I_j by δ_j, four of them cover it.

First let $I = (a, b]$ be bounded, so all $\mu_0(I_j)$ are finite, and let $\varepsilon > 0$. Choose $\delta > 0$ with $F(a + \delta) - F(a) < \varepsilon$, and for each j choose $\delta_j > 0$ with $F(b_j + \delta_j) - F(b_j) < \varepsilon/2^j$. The open intervals $(a_j, b_j + \delta_j)$ cover the compact interval $[a + \delta, b]$, so finitely many of them do. Take a minimal finite subcover and relabel it as $(a_1, b_1 + \delta_1), \ldots, (a_N, b_N + \delta_N)$, ordered by left endpoints. Consecutive intervals then overlap (otherwise a point of $[a + \delta, b]$ between them would be missed, or one of them could be dropped), so

$$a_1 < a + \delta, \qquad a_{j+1} < b_j + \delta_j \quad (1 \le j < N), \qquad b < b_N + \delta_N.$$

Since F is increasing,

$$\begin{aligned} F(b) - F(a + \delta) &\le F(b_N + \delta_N) - F(a_1) \\ &= \big[F(b_N + \delta_N) - F(a_N)\big] + \sum_{j=1}^{N-1} \big[F(a_{j+1}) - F(a_j)\big] \\ &\le \sum_{j=1}^{N} \big[F(b_j + \delta_j) - F(a_j)\big] \\ &= \sum_{j=1}^{N} \big[F(b_j) - F(a_j)\big] + \sum_{j=1}^{N} \big[F(b_j + \delta_j) - F(b_j)\big] \;<\; \sum_{j=1}^{\infty} \mu_0(I_j) + \varepsilon, \end{aligned}$$

since the chosen intervals are distinct and all the lengthenings together cost less than $\sum_j \varepsilon/2^j = \varepsilon$. Hence

$$\mu_0(I) = F(b) - F(a) < F(b) - F(a + \delta) + \varepsilon < \sum_{j=1}^{\infty} \mu_0(I_j) + 2\varepsilon,$$

and letting $\varepsilon \to 0$ gives "$\le$".

If I is unbounded, apply the bounded case to $I \cap (-M, M]$, the disjoint union of the h-intervals $I_j \cap (-M, M]$. Since F is increasing, $\mu_0(I_j \cap (-M, M]) \le \mu_0(I_j)$, so $\mu_0(I \cap (-M, M]) \le \sum_j \mu_0(I_j)$. As $M \to \infty$, $\mu_0(I \cap (-M, M]) \to \mu_0(I)$ by the definition of $F(\pm\infty)$.

Explanation. Right continuity cannot be dropped. $F := 0$ on $(-\infty, 0]$ and $F := 1$ on $(0, \infty)$ is increasing but not right continuous at 0, and countable additivity fails: $(0, 1]$ is the disjoint union of the h-intervals $(\frac{1}{k+1}, \frac{1}{k}]$, $k \ge 1$, each with $\mu_0 = 0$, while $\mu_0((0, 1]) = F(1) - F(0) = 1$. In step (iv), no $\delta > 0$ has $F(\delta) - F(0) < \varepsilon$ once $\varepsilon < 1$.

8.5 The Lebesgue–Stieltjes Theorem

The Proposition above (an increasing, right continuous F defines a premeasure μ_0 on $\mathcal{A}$), together with Carathéodory's theorem, gives a large class of explicit and important measures on $\mathbb{R}$:

Theorem

($\Rightarrow$) Let $F : \mathbb{R} \to \mathbb{R}$ be increasing and right continuous. Then there is a unique Borel measure μ_F on $\mathbb{R}$, the *Lebesgue–Stieltjes measure* of F, with

$$\mu_F((a, b]) = F(b) - F(a) \qquad \text{for all } a < b.$$

If G is another such function, then $\mu_F = \mu_G$ iff $F - G$ is constant.

($\Leftarrow$) Conversely, let μ be a Borel measure on $\mathbb{R}$ that is finite on all bounded Borel sets, and set

$$F(x) := \begin{cases} \mu((0, x]), & x > 0, \\ 0, & x = 0, \\ -\mu((x, 0]), & x < 0 \end{cases}$$

(the distribution function of μ). Then F is increasing and right continuous, and $\mu = \mu_F$.

Proof

($\Rightarrow$) By the Proposition, F induces a premeasure μ_0 on $\mathcal{A}$. By Carathéodory, μ_0 extends to a measure μ_F on the σ-algebra generated by $\mathcal{A}$, which is $\mathcal{B}_\mathbb{R}$ (the h-intervals generate $\mathcal{B}_\mathbb{R}$). Moreover, μ_0 is σ-finite:

$$\mathbb{R} = \bigcup_{j \in \mathbb{Z}} (j, j+1], \qquad \mu_0((j, j+1]) = F(j+1) - F(j) < \infty.$$

So μ_F is unique, by uniqueness in Carathéodory: any Borel measure ν with $\nu((a, b]) = F(b) - F(a)$ for all $a < b$ agrees with μ_0 on $\mathcal{A}$ by additivity, hence $\nu = \mu_F$.

If $G = F + c$ with c constant, then $\mu_G = \mu_F$, since the premeasures are the same:

$$\mu_G((a, b]) = (F + c)(b) - (F + c)(a) = F(b) - F(a) = \mu_F((a, b]).$$

Conversely, if $\mu_F = \mu_G$, then $F(b) - F(a) = G(b) - G(a)$ for all $a < b$, i.e. $F - G$ is constant.

($\Leftarrow$) Let μ be a Borel measure, finite on bounded Borel sets, and define F as above. As in the Claim on distribution functions, F is increasing and right continuous; at $x < 0$, where $F(x) = -\mu((x, 0])$, use continuity from below instead of from above.

Claim: $\mu|_\mathcal{A} = \mu_F|_\mathcal{A}$. Since both are additive, it is enough to check this on h-intervals. E.g. for $(a, b]$ with $0 < a < b$, additivity of μ (and $\mu((0, a]) < \infty$) gives

$$\mu((a, b]) = \mu((0, b]) - \mu((0, a]) = F(b) - F(a) = \mu_F((a, b]).$$

The other cases are similar; e.g. for $a < 0 < b$, $\mu((a, b]) = \mu((a, 0]) + \mu((0, b]) = -F(a) + F(b)$. Hence $\mu|_\mathcal{A} = \mu_F|_\mathcal{A}$, and by uniqueness in Carathéodory (μ_0 is σ-finite), $\mu = \mu_F$ on $\mathcal{B}_\mathbb{R}$.

Remark. By the completion theorem (Section 6), μ_F extends uniquely to a complete measure $\overline{\mu_F}$ on the completion $\mathcal{M}_\mu := \overline{\mathcal{B}_\mathbb{R}}$. As agreed in Section 6, we work with these completions, and, abusing

notation, we call the completion just μ_F.

Example

Take $F(x) = x$. Then $\overline{\mu_F}$ is the Lebesgue measure m on $\mathcal{L} := \mathcal{L}_{\mathbb{R}} = \overline{\mathcal{B}_{\mathbb{R}}}$ of Section 7 (case $n = 1$): both measures give $(a, b]$ the value $b - a$, so they agree on $\mathcal{B}_{\mathbb{R}}$ by uniqueness, hence also on the completion.

Explanation. Think of F as recording cumulative mass: μ_F gives $(a, b]$ the mass $F(b) - F(a)$. Where F is constant there is no mass, and a jump of F at x is a point mass, $\mu_F(\{x\}) = F(x) - F(x^-)$. For example, $F = 0$ on $(-\infty, 0)$ and $F = 1$ on $[0, \infty)$ gives the delta measure δ_0 of Section 5. The normalization $F(0) = 0$ in the converse is used because $\mu((-\infty, x])$ may be infinite, as it is for m.

8.6 Regularity

Question. How do we estimate $\mu_F(E)$, $E \in \mathcal{M}_\mu$, in terms of "simple" sets (open, closed, compact)?

Note. Let $\mu = \mu_F$ and $E \in \mathcal{M}_\mu$. Then $\mu(E) = \mu^*(E)$, so

$$\mu(E) = \inf\Big\{ \sum_{j=1}^{\infty} \mu((a_j, b_j]) \;\Big|\; E \subseteq \bigcup_{j=1}^{\infty} (a_j, b_j] \Big\}$$
$$= \inf\Big\{ \sum_{j=1}^{\infty} \big[F(b_j) - F(a_j)\big] \;\Big|\; E \subseteq \bigcup_{j=1}^{\infty} (a_j, b_j] \Big\}. \qquad (*)$$

Here covers by sets in $\mathcal{A}$ may be replaced by covers by h-intervals, since every set in $\mathcal{A}$ is a finite disjoint union of h-intervals.

The first observation is that in $(*)$ one can replace h-intervals by open intervals.

Lemma

Let $E \in \mathcal{M}_\mu$. Then

$$\mu(E) = \inf\Big\{ \sum_{j=1}^{\infty} \mu((a_j, b_j)) \;\Big|\; E \subseteq \bigcup_{j=1}^{\infty} (a_j, b_j) \Big\}.$$

Proof

Let $\nu(E)$ denote the right-hand side.

$(\nu(E) \geq \mu(E))$ Suppose $E \subseteq \bigcup_{j=1}^{\infty} (a_j, b_j)$. Each open interval is a countable disjoint union of h-intervals, $(a_j, b_j) = \bigcup_{k=1}^{\infty} I_j^k$; e.g. $I_j^k = (c_{k-1}, c_k]$ with $c_k := b_j - 2^{-k}(b_j - a_j)$. Then $E \subseteq \bigcup_{j,k} I_j^k$ is a cover by h-intervals, so

$$\sum_{j=1}^{\infty} \mu((a_j, b_j)) = \sum_{j,k=1}^{\infty} \mu(I_j^k) \geq \mu^*(E) = \mu(E).$$

This holds for every such cover, so $\nu(E) \geq \mu(E)$.

($\nu(E) \le \mu(E)$) Here we use the right continuity of F. Let $\varepsilon > 0$. By the definition of $\mu(E) = \mu^*(E)$ there are h-intervals $(a_j, b_j]$ with

ⓐ $E \subseteq \bigcup_{j=1}^{\infty}(a_j, b_j]$,
ⓑ $\sum_{j=1}^{\infty} \mu((a_j, b_j]) \le \mu(E) + \varepsilon$.

By right continuity of F there are $\delta_j > 0$ with $F(b_j + \delta_j) - F(b_j) < \varepsilon/2^j$. Then

$$E \subseteq \bigcup_{j=1}^{\infty}(a_j, b_j] \subseteq \bigcup_{j=1}^{\infty}(a_j, b_j + \delta_j),$$

and since $\mu((a, b)) \le \mu((a, b]) = F(b) - F(a)$,

$$\begin{aligned}
\sum_{j=1}^{\infty} \mu((a_j, b_j + \delta_j)) &\le \sum_{j=1}^{\infty} \big[F(b_j + \delta_j) - F(a_j)\big] \\
&= \sum_{j=1}^{\infty} \big[F(b_j + \delta_j) - F(b_j)\big] + \sum_{j=1}^{\infty} \big[F(b_j) - F(a_j)\big] \\
&\le \sum_{j=1}^{\infty} \frac{\varepsilon}{2^j} + \sum_{j=1}^{\infty} \mu((a_j, b_j]) \\
&\le \varepsilon + \mu(E) + \varepsilon = \mu(E) + 2\varepsilon.
\end{aligned}$$

Since $\varepsilon > 0$ is arbitrary, $\nu(E) \le \mu(E)$. Hence $\mu(E) = \nu(E)$.

The following is a natural extension of the Lemma.

Theorem (1)

Let $\mu = \mu_F$ and $E \in \mathcal{M}_\mu$. Then

$$\begin{aligned}
\mu(E) &= \inf\{\mu(U) \mid U \supseteq E,\ U \text{ open}\} && (\dagger) \\
&= \sup\{\mu(K) \mid K \subseteq E,\ K \text{ compact}\}. && (\dagger\dagger)
\end{aligned}$$

Proof (†)

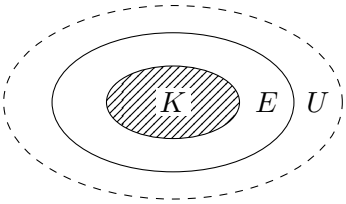
Let $\varepsilon > 0$. By the Lemma there are open intervals (a_j, b_j) with

① $E \subseteq \bigcup_{j=1}^{\infty}(a_j, b_j)$,
② $\sum_{j=1}^{\infty} \mu((a_j, b_j)) \le \mu(E) + \varepsilon$.

Then $U := \bigcup_{j=1}^{\infty}(a_j, b_j)$ is open, $U \supseteq E$, and by subadditivity $\mu(U) \le \sum_j \mu((a_j, b_j)) \le \mu(E) + \varepsilon$. So the infimum in (†) is at most $\mu(E) + \varepsilon$, and since $\varepsilon > 0$ is arbitrary, it is at most $\mu(E)$. Conversely, $E \subseteq U$ implies $\mu(E) \le \mu(U)$ for every open $U \supseteq E$, by monotonicity.

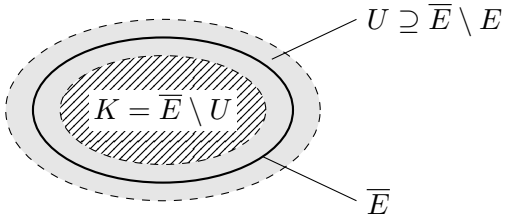
9 Regularity, Lebesgue Measure and the Cantor Set

It remains to prove (††) in Theorem 1, approximation from inside. The picture: $K \subseteq E \subseteq U$.



Proof (††)

Case 1: E bounded. If E is closed, then E is compact and (††) is trivial (take $K = E$). So assume $\overline{E} \setminus E \neq \emptyset$. Clearly $\overline{E} \setminus E \in \mathcal{M}_\mu$. Also $\mu(\overline{E}) < \infty$: $\overline{E}$ is bounded, so $\overline{E} \subseteq (a, b]$ for some $a < b$, and $\mu((a, b]) = F(b) - F(a) < \infty$.



Let $\varepsilon > 0$. By (†) there is an open $U \supseteq \overline{E} \setminus E$ with

$$\mu(U) \leq \mu(\overline{E} \setminus E) + \varepsilon. \tag{1}$$

Set $K := \overline{E} \setminus U$. It is closed and bounded, hence compact, and $K \subseteq E$ because $U \supseteq \overline{E} \setminus E$; in fact $K = E \setminus U$. All the measures below are finite ($\mu(E) \leq \mu(\overline{E}) < \infty$, and $\mu(U) < \infty$ by (1)), so

$$\begin{aligned}
\mu(K) &= \mu(E) - \mu(E \cap U) \\
&= \mu(E) - \big[\mu(U) - \mu(U \setminus E)\big] \\
&= \mu(E) - \mu(U) + \mu(U \setminus E) \\
&\geq \mu(E) - \mu(U) + \mu(\overline{E} \setminus E) && \text{since } U \setminus E \supseteq \overline{E} \setminus E \\
&\geq \mu(E) - \big[\mu(\overline{E} \setminus E) + \varepsilon\big] + \mu(\overline{E} \setminus E) && \text{by (1)} \\
&= \mu(E) - \varepsilon.
\end{aligned}$$

By monotonicity also $\mu(K) \leq \mu(E)$, since $K \subseteq E$. As $\varepsilon > 0$ is arbitrary, $\sup\{\mu(K) \mid K \subseteq E \text{ compact}\} = \mu(E)$, provided E is bounded.

Case 2: general E. Let $E_j := E \cap (j, j+1]$, $j \in \mathbb{Z}$. Each E_j is bounded, so by Case 1 there are compact sets K_j with

① $K_j \subseteq E_j$,
② $\mu(K_j) \geq \mu(E_j) - \varepsilon/2^{|j|+2}$.

To approximate E as well as possible from inside, consider the compact sets

$$H_n := \bigcup_{j=-n}^{n} K_j \subseteq E, \qquad n = 1, 2, \ldots$$

The K_j are disjoint (as the E_j are), and $\sum_{j\in\mathbb{Z}} 2^{-|j|-2} = \frac{1}{4} + 2 \cdot \frac{1}{4} = \frac{3}{4}$, so

$$\mu(H_n) = \sum_{j=-n}^{n} \mu(K_j) \geq \sum_{j=-n}^{n} \mu(E_j) - \varepsilon = \mu\Big(\bigcup_{j=-n}^{n} E_j\Big) - \varepsilon. \tag{2}$$

The H_n increase, and $\bigcup_{j=-n}^{n} E_j \nearrow \bigcup_{j\in\mathbb{Z}} E_j = E$, so continuity of measure and (2) give

$$\lim_n \mu(H_n) \geq \lim_n \mu\Big(\bigcup_{j=-n}^{n} E_j\Big) - \varepsilon = \mu(E) - \varepsilon,$$

where both sides are $+\infty$ if $\mu(E) = \infty$. Each H_n is a compact subset of E, so $\sup\{\mu(K) \mid K \subseteq E \text{ compact}\} \geq \lim_n \mu(H_n) \geq \mu(E) - \varepsilon$ for every $\varepsilon > 0$. Together with $\mu(K) \leq \mu(E)$ for $K \subseteq E$, this proves (††) for every $E \in \mathcal{M}_\mu$.

Upshot. Given $E \in \mathcal{M}_\mu$, we can approximate $\mu(E)$ arbitrarily well from *outside* by open sets and arbitrarily well from *inside* by compact sets.

Explanation. Property (†) is called *outer regularity* and (††) *inner regularity*.

9.1 G_δ and F_σ Characterization

The following result characterizes the sets $E \in \mathcal{M}_\mu$ by G_δ and F_σ sets, modulo measure zero.

Theorem (2)

Let $\mu = \mu_F$ and $E \subseteq \mathbb{R}$. The following are equivalent:

ⓐ $E \in \mathcal{M}_\mu$;
ⓑ $E = V \setminus N_1$ with V a G_δ set and $\mu(N_1) = 0$;
ⓒ $E = H \cup N_2$ with H an F_σ set and $\mu(N_2) = 0$.

Proof

ⓑ or ⓒ $\implies$ ⓐ: G_δ and F_σ sets are Borel, so they lie in $\mathcal{B}_\mathbb{R} \subseteq \mathcal{M}_\mu$, and null sets lie in $\mathcal{M}_\mu$ since it is complete. Hence $E \in \mathcal{M}_\mu$.

ⓐ $\implies$ ⓑ and ⓒ: Suppose $\mu(E) < \infty$. By Theorem 1 (with $\varepsilon = 2^{-j}$) there are open sets $U_j \supseteq E$ and compact, hence closed, sets $K_j \subseteq E$ with

$$\mu(U_j) - 2^{-j} \leq \mu(E) \leq \mu(K_j) + 2^{-j}, \qquad j = 1, 2, \ldots \tag{$*$}$$

Set

$$V := \bigcap_{j=1}^{\infty} U_j \quad (\text{a } G_\delta \text{ set}), \qquad H := \bigcup_{j=1}^{\infty} K_j \quad (\text{an } F_\sigma \text{ set}).$$

Then $H \subseteq E \subseteq V$. Moreover, by continuity of measure and $(*)$,

$$\mu(H) = \lim_n \mu\Big(\bigcup_{j=1}^{n} K_j\Big) \geq \mu(E), \qquad \text{since } \mu\Big(\bigcup_{j=1}^{n} K_j\Big) \geq \mu(K_n) \geq \mu(E) - 2^{-n}.$$

On the other hand $\mu(H) \leq \mu(E)$ since $H \subseteq E$, so $\mu(H) = \mu(E)$. Similarly, $\mu(E) \leq \mu(V) \leq \mu(U_n) \leq \mu(E) + 2^{-n}$ for all n, so $\mu(V) = \mu(E)$. Therefore $\mu(V \setminus E) = \mu(V) - \mu(E) = 0$ and $\mu(E \setminus H) = \mu(E) - \mu(H) = 0$, as $\mu(E) < \infty$, so ⓑ holds with $N_1 := V \setminus E$ and ⓒ holds with $N_2 := E \setminus H$.

Exercise

Complete the proof for general $E \in \mathcal{M}_\mu$, where $\mu(E) = \infty$ is allowed. Use that μ is σ-finite: $E = \bigcup_{j\in\mathbb{Z}} \big(E \cap (j, j+1]\big)$, and each piece has finite measure. (Careful: a countable union of G_δ sets need not be a G_δ set.)

Upshot. $\mathcal{M}_\mu$ is quite simple modulo measure zero.

Explanation. The null sets cannot be dropped. For Lebesgue measure, $\mathbb{Q}$ is F_σ (a countable union of points) but, by the Baire category theorem, not G_δ; so in (b) the G_δ set V must be strictly larger than $\mathbb{Q}$, e.g. a G_δ set of measure zero containing $\mathbb{Q}$. Dually, $\mathbb{R} \setminus \mathbb{Q}$ is G_δ but not F_σ, so in (c) the set N_2 cannot be empty.

9.2 Approximation by Finite Unions of Intervals

There is another approximation result that is quite useful:

Proposition

Let $E \in \mathcal{M}_\mu$ with $\mu(E) < \infty$. Then for every $\varepsilon > 0$ there is a finite union of open intervals $A = \bigcup_{j=1}^n (a_j, b_j)$ such that

$$\mu(E \triangle A) < \varepsilon.$$

Proof

The lecture left this as an exercise. Let $\varepsilon > 0$. By the Lemma before Theorem 1 (Section 8) there are open intervals (a_j, b_j), $j \in \mathbb{N}$, with

$$E \subseteq U := \bigcup_{j=1}^{\infty} (a_j, b_j) \qquad \text{and} \qquad \sum_{j=1}^{\infty} \mu((a_j, b_j)) < \mu(E) + \frac{\varepsilon}{2}.$$

The series converges, so there is n with $\sum_{j>n} \mu((a_j, b_j)) < \frac{\varepsilon}{2}$. Let $A := \bigcup_{j=1}^n (a_j, b_j)$. Then:

- $A \setminus E \subseteq U \setminus E$, and $\mu(U \setminus E) = \mu(U) - \mu(E) < \frac{\varepsilon}{2}$, since $\mu(U) \leq \sum_j \mu((a_j, b_j))$ by subadditivity and $\mu(E) < \infty$;
- $E \setminus A \subseteq U \setminus A \subseteq \bigcup_{j>n} (a_j, b_j)$, so $\mu(E \setminus A) \leq \sum_{j>n} \mu((a_j, b_j)) < \frac{\varepsilon}{2}$.

Hence $\mu(E \triangle A) = \mu(A \setminus E) + \mu(E \setminus A) < \varepsilon$.

Lebesgue measure m on $\mathcal{L}$ has all the usual properties of measures, but it also has important invariance properties which are special to the Lebesgue case.

9.3 Invariance under Translations and Dilations

Theorem

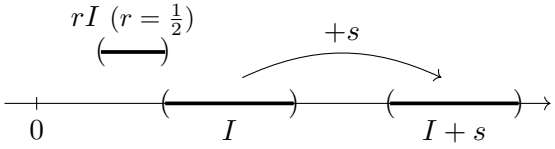
Let $E \in \mathcal{L}$, $s \in \mathbb{R}$ and $r \in \mathbb{R} \setminus \{0\}$. Then $E + s \in \mathcal{L}$ and $rE \in \mathcal{L}$, where

$$E + s := \{x + s \mid x \in E\}, \qquad rE := \{rx \mid x \in E\}.$$

Moreover, $m(E + s) = m(E)$ and $m(rE) = |r|\, m(E)$.

Proof

Let $I \subseteq \mathbb{R}$ be an open interval. Then $I + s$ and rI are open intervals (here $r \neq 0$ is needed).



Since open intervals generate $\mathcal{B}_\mathbb{R}$, it follows that $\mathcal{B}_\mathbb{R}$ is invariant under translations and dilations, i.e. $E \in \mathcal{B}_\mathbb{R} \implies E + s,\ rE \in \mathcal{B}_\mathbb{R}$: the sets E with $E + s \in \mathcal{B}_\mathbb{R}$ form a σ-algebra containing the open intervals, hence all of $\mathcal{B}_\mathbb{R}$, and likewise for rE.

For $E \in \mathcal{B}_\mathbb{R}$ set

$$m_s(E) := m(E + s), \qquad m^r(E) := m(rE);$$

these are again measures on $\mathcal{B}_\mathbb{R}$. We want to show that $m(E + s) = m(E)$ and $m(rE) = |r|\, m(E)$ for all $E \in \mathcal{L}$, starting with $E \in \mathcal{B}_\mathbb{R}$. Let $\mathcal{F}$ be the family of finite unions of intervals. Clearly

$$m_s(E) = m(E), \qquad m^r(E) = |r|\, m(E) \qquad \text{for all } E \in \mathcal{F}. \tag{$*$}$$

In particular $(*)$ holds on $\mathcal{A} \subseteq \mathcal{F}$. So m_s and m are measures on $\mathcal{B}_\mathbb{R} = \mathcal{M}(\mathcal{A})$ that agree on $\mathcal{A}$, where m is σ-finite, and uniqueness in Carathéodory II gives $m_s = m$ on $\mathcal{B}_\mathbb{R}$; similarly $m^r = |r|\, m$ on $\mathcal{B}_\mathbb{R}$. So m is invariant on $\mathcal{B}_\mathbb{R}$. In particular, $\{E \in \mathcal{B}_\mathbb{R} \mid m(E) = 0\}$ is invariant under translation and scaling.

Measure zero. Every $E \in \mathcal{L}$ has the form $E = B \cup Z$ with $B \in \mathcal{B}_\mathbb{R}$ (invariant, as just shown) and $Z \subseteq N$ for some $N \in \mathcal{B}_\mathbb{R}$ with $m(N) = 0$. Given any $Z \in \mathcal{L}$ with $m(Z) = 0$, we get $Z + s,\ rZ \in \mathcal{L}$ and $m(Z + s) = m(rZ) = 0$: Z lies in a Borel set N with $m(N) = 0$, so $Z + s \subseteq N + s$ and $rZ \subseteq rN$, which are Borel null sets by the Borel case, and m is complete on $\mathcal{L}$. Hence $E + s = (B + s) \cup (Z + s) \in \mathcal{L}$, $rE = rB \cup rZ \in \mathcal{L}$, and

$$m(E + s) = m(B + s) = m(B) = m(E), \qquad m(rE) = m(rB) = |r|\, m(B) = |r|\, m(E).$$

Explanation. The factor is $|r|$, not r: dilation by $r < 0$ reverses each interval, e.g. $-(a, b] = [-b, -a)$, but still multiplies its length by $|r|$. The theorem also holds for $r = 0$, since $0E \subseteq \{0\}$ gives $m(0E) = 0 = 0 \cdot m(E)$ (with $0 \cdot \infty = 0$); only the proof needs $r \neq 0$.

9.4 The Cantor Set

Fact

Let F be increasing and continuous, with exterior measure μ_F^*. Then every countable set $S \subseteq \mathbb{R}$ has $\mu_F^*(S) = 0$.

> ***Proof***
>
> For $x \in \mathbb{R}$ and $\delta > 0$, $\{x\} \subseteq (x - \delta, x]$, so $\mu_F^*(\{x\}) \leq F(x) - F(x - \delta)$, which tends to 0 as $\delta \to 0$ since F is continuous. So points have outer measure 0, and countable subadditivity gives $\mu_F^*(S) = 0$.

In particular, for Lebesgue measure ($F(x) = x$): $m(S) = 0$ for every countable $S \subseteq \mathbb{R}$.

Question. What about the converse, i.e. if $m(S) = 0$, is S countable?

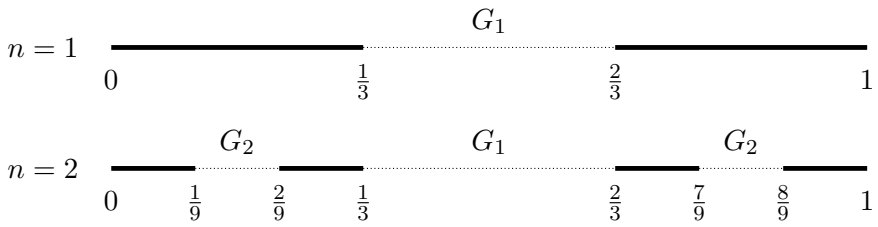
Answer. No: the Cantor set.

Definition

The *Cantor set* is

$$\mathcal{C} := [0,1] \setminus \bigcup_{n=1}^{\infty} G_n,$$

where the open sets G_n are given as follows: $G_1 := (\frac{1}{3}, \frac{2}{3})$, and G_n is the union of the open middle thirds of the intervals of $[0,1] \setminus \bigcup_{j=1}^{n-1} G_j$.



Note. $\mathcal{C} \neq \emptyset$, since the endpoints of the intervals in the construction lie in $\mathcal{C}$; e.g. $\frac{1}{3} \in \mathcal{C}$. In fact $\mathcal{C}$ also contains many points that are not endpoints.

9.5 Base-3 Expansions

The Cantor set has a nice description in terms of base-3 expansions of numbers in $[0,1]$. Write

$$x = \sum_{n=1}^{\infty} \frac{a_n}{3^n} =: 0.a_1a_2a_3\ldots_3, \qquad a_n \in \{0,1,2\}.$$

Then

$$G_1 = \{x \in [0,1] \mid \text{every base-3 expansion of } x \text{ has } a_1 = 1\},$$
$$G_1 \cup G_2 = \{x \in [0,1] \mid \text{every base-3 expansion of } x \text{ has } a_1 = 1 \text{ or } a_2 = 1\},$$

etc. Hence

$$\bigcup_{n=1}^{\infty} G_n = \{x \in [0,1] \mid \text{every base-3 expansion of } x \text{ contains a } 1\},$$

and therefore

$$\mathcal{C} = [0,1] \setminus \bigcup_{n=1}^{\infty} G_n = \{x \in [0,1] \mid x \text{ has some base-3 expansion with no 1's}\}.$$

Example

$x = \frac{1}{3}$ has two base-3 expansions, $0.1000\ldots_3$ and $0.0222\ldots_3$, so expansions are not unique. Only the second has no 1's, which is why the description of $\mathcal{C}$ says *some* expansion.

Explanation. The n-th base-3 digit of x says which third of the current interval x lies in (0: left, 1: middle, 2: right), so the first digit 1 means that x falls into a removed middle third or onto one of its endpoints. Only numbers of the form $k/3^n$ have two expansions, and for the endpoints of the removed intervals one of the two avoids 1's, e.g. $\frac{2}{3} = 0.1222\ldots_3 = 0.2000\ldots_3$. A point of $\mathcal{C}$ that is not an endpoint is $\frac{1}{4} = 0.020202\ldots_3$.

Proposition

The Cantor set has the following properties:

ⓐ $\mathcal{C}$ is closed;
ⓑ $m(\mathcal{C}) = 0$;
ⓒ $\mathcal{C}$ contains no non-trivial interval (a,b), $a < b$.

Proof

ⓐ $\bigcup_{n=1}^{\infty} G_n$ is open, since the G_n are open. Hence

$$\mathcal{C} = [0,1] \cap \Big(\mathbb{R} \setminus \bigcup_{n=1}^{\infty} G_n\Big)$$

is an intersection of two closed sets, so $\mathcal{C}$ is closed.

ⓑ G_n is a union of 2^{n-1} disjoint open intervals of length 3^{-n}: each step removes the middle third of every remaining interval, doubling their number and dividing their length by 3. So $m(G_n) = 2^{n-1}/3^n$, and since the G_n are disjoint,

$$m\Big(\bigcup_{n=1}^{\infty} G_n\Big) = \sum_{n=1}^{\infty} \frac{2^{n-1}}{3^n} = \frac{1}{3}\sum_{n=1}^{\infty}\Big(\frac{2}{3}\Big)^{n-1} = \frac{1}{3}\cdot\frac{1}{1-2/3} = 1.$$

Therefore $m(\mathcal{C}) = m([0,1]) - m\big(\bigcup_{n=1}^{\infty} G_n\big) = 1 - 1 = 0$.

ⓒ If $(a,b) \subseteq \mathcal{C}$ with $a < b$, then by ⓑ, $0 = m(\mathcal{C}) \geq m((a,b)) = b - a > 0$, a contradiction.

Explanation. Directly: $\mathcal{C}$ lies in the set $[0,1] \setminus \bigcup_{k=1}^{n} G_k$ left after n steps, which has total length $(2/3)^n$, so $m(\mathcal{C}) \leq (2/3)^n \to 0$.

9.6 The Cantor Function

Definition

The *Cantor function* $f : [0,1] \to [0,1]$ converts base 3 to base 2 as follows.

① If $x \in \mathcal{C}$, write $x = \sum_{n=1}^{\infty} \frac{2a_n}{3^n}$ with $a_n \in \{0,1\}$ (the base-3 expansion of x with no 1's, which is unique), and set

$$f(x) := \sum_{n=1}^{\infty} \frac{a_n}{2^n} \qquad \text{(base 2)}.$$

② If $x \notin \mathcal{C}$, truncate the base-3 expansion of x after its first 1 and replace the 2's before it by 1's (any expansion of x gives the same result). Read the resulting string of 0's and 1's as a base-2 expansion; this is $f(x)$.

Example

Two values of f:

$$f(0.0202_3) = 0.0101_2, \qquad \text{i.e. } f\left(\tfrac{20}{81}\right) = \tfrac{5}{16} \qquad (0.0202_3 \in \mathcal{C});$$
$$f(0.220121_3) = 0.1101_2, \qquad \text{i.e. } f\left(\tfrac{664}{729}\right) = \tfrac{13}{16} = \tfrac{1}{2} + \tfrac{1}{4} + \tfrac{1}{16} \qquad (0.220121_3 \notin \mathcal{C}).$$

In the second example the first 1 is the fourth digit: truncating gives 0.2201_3, and replacing the 2's by 1's gives 0.1101_2.

Exercise

Graph f on $\bigcup_{k=1}^{n} G_k$. (The lecture has $[0,1] \setminus \bigcup_{k=1}^{n} G_k$, apparently a slip: on $\bigcup_{k=1}^{n} G_k$, f is a step function.)

Explanation. On $\mathcal{C}$, f keeps the digit pattern and only changes the base, so it matches $\mathcal{C}$ with $[0,1]$ almost one-to-one: the only collisions are the two endpoints of each removed interval, which are sent to the same point, e.g. $f(\tfrac{1}{3}) = f(0.0222\ldots_3) = 0.0111\ldots_2 = \tfrac{1}{2} = f(0.2_3) = f(\tfrac{2}{3})$.

Proposition

The Cantor function $f : [0,1] \to [0,1]$ is increasing and onto $[0,1]$, and $f(\mathcal{C}) = [0,1]$. Moreover, $f \in C^0([0,1])$.

Corollary

$\mathcal{C}$ is uncountable, since its image $f(\mathcal{C}) = [0,1]$ is uncountable.

The continuity of f will follow from a general fact:

Lemma

Suppose $f : [0,1] \to [0,1]$ is increasing and $f([0,1]) = [0,1]$. Then $f \in C^0([0,1])$.

Proof

Exercise. (*Hint:* at a discontinuity, an increasing function jumps over an interval of values.)

Proof (of Prop.)

To show $f(\mathcal{C}) = [0,1]$, let $y \in [0,1]$, $y = \sum_{n=1}^{\infty} a_n/2^n$ with $a_n \in \{0,1\}$. Replacing each digit $a_n = 1$ by 2 gives the base-3 expansion of a point $x \in [0,1]$,

$$x = \sum_{n=1}^{\infty} \frac{b_n}{3^n}, \qquad b_n = 2a_n \in \{0,2\}.$$

So $x \in \mathcal{C}$, and $f(x) = y$ by part ① of the definition of f. Hence $f(\mathcal{C}) = [0,1]$, and in particular f is onto.

Note that if $x \in [0,1] \setminus \mathcal{C}$, then f is constant on any open interval $I_x \subseteq [0,1] \setminus \mathcal{C}$ containing x: such an interval lies in a single removed interval, whose points all have the same base-3 digits up to their first 1, so part ② of the definition gives them the same value. Moreover, f is increasing (Exercise; compare base-3 expansions digit by digit). Finally, applying the Lemma to the Cantor function gives $f \in C^0([0,1])$.

9.7 A Null Set with Non-measurable Image

Proposition

Let f be the Cantor function. There is a set $A \subseteq [0,1]$ with $A \in \mathcal{L}$ and $m(A) = 0$, but $f(A) \notin \mathcal{L}$.

Proof

Exercise. (*Hint:* use the existence of a set $B \subseteq [0,1]$ with $B \notin \mathcal{L}$.)

Explanation. So even a continuous increasing function can destroy measurability: A is as small as possible in measure, yet its image is not in $\mathcal{L}$ at all. This is one reason why measurability of functions will be defined through preimages rather than images (Section 10).

10 Measurable Functions

10.1 Motivation and Requirements

So far we have studied measurable sets and σ-algebras $\mathcal{M}$. Next, given $E \in \mathcal{M}$, we want to define the integral $\int_E f\,d\mu$ for a "large" class of functions $f : X \to \mathbb{R}$.

Remark. For $\mu = m$ (Lebesgue measure) and $\mathcal{M} = \mathcal{L}$, $\int_E f\,dm$ is the *Lebesgue integral*.

Question. Which functions $f : X \to \mathbb{R}$ (or $\mathbb{C}$) should we integrate?

Requirements. The class should:

① contain the "well-behaved" (e.g. continuous) functions;

② behave well with respect to taking limits.

10.2 Measurable Maps

Let $(X, \mathcal{M})$ be a measurable space, $f : X \to Y$ a map, and $\mathcal{N}$ a σ-algebra on Y.

Note. $f^{-1}(\mathcal{N}) := \{f^{-1}(E) \mid E \in \mathcal{N}\}$ is a σ-algebra on X, the *pull-back* of $\mathcal{N}$. Indeed $f^{-1}(Y) = X$, and preimages commute with complements and countable unions (basic relations, Section 2).

Definition

$f : X \to Y$ is $(\mathcal{M}, \mathcal{N})$-*measurable* if $f^{-1}(E) \in \mathcal{M}$ for all $E \in \mathcal{N}$.

Note. This is similar to the definition of a continuous map ($f^{-1}(U)$ open in X for all open $U \subseteq Y$), but much more general.

Note. Compositions of measurable maps are measurable. Let $g : (X, \mathcal{M}) \to (Y, \mathcal{N})$ and $f : (Y, \mathcal{N}) \to (Z, \mathcal{O})$ be measurable. For $E \in \mathcal{O}$,

$$(f \circ g)^{-1}(E) = g^{-1}\big(f^{-1}(E)\big),$$

where $f^{-1}(E) \in \mathcal{N}$, so $g^{-1}(f^{-1}(E)) \in \mathcal{M}$. Hence $f \circ g : X \to Z$ is $(\mathcal{M}, \mathcal{O})$-measurable.

Explanation. For $f : X \to \mathbb{R}$ (with $\mathcal{N} = \mathcal{B}_{\mathbb{R}}$), measurability says that sets such as $\{x \in X \mid a < f(x) \leq b\}$ lie in $\mathcal{M}$, so a measure on $\mathcal{M}$ can assign them a size. The integral of $f \geq 0$ will be built from the sizes of exactly such sets (approximation theorem, Section 11). In terms of the pull-back: f is $(\mathcal{M}, \mathcal{N})$-measurable iff $f^{-1}(\mathcal{N}) \subseteq \mathcal{M}$, so $f^{-1}(\mathcal{N})$ is the smallest σ-algebra on X for which f is measurable.

10.3 Checking on Generating Sets

It suffices to check measurability on a generating family. Write $\langle \mathcal{E} \rangle$ for the σ-algebra generated by $\mathcal{E}$ (denoted $\mathcal{M}(\mathcal{E})$ in Section 3).

Proposition

Let $f : X \to Y$, and let $\mathcal{N} = \langle \mathcal{E} \rangle$ with $\mathcal{E} \subseteq \mathcal{P}(Y)$. Then f is $(\mathcal{M}, \mathcal{N})$-measurable iff $f^{-1}(E) \in \mathcal{M}$ for all $E \in \mathcal{E}$.

> ***Proof***
>
> ($\Rightarrow$) $E \in \mathcal{E} \implies E \in \mathcal{N} \implies f^{-1}(E) \in \mathcal{M}$.
>
> ($\Leftarrow$) The collection $\mathcal{N}^* := \{E \subseteq Y \mid f^{-1}(E) \in \mathcal{M}\}$ is a σ-algebra on Y, since preimages commute with complements and countable unions (as for $\mathcal{N}_\alpha$ in Section 4). By assumption $\mathcal{E} \subseteq \mathcal{N}^*$, so $\mathcal{N} = \langle \mathcal{E} \rangle \subseteq \mathcal{N}^*$, i.e. f is $(\mathcal{M}, \mathcal{N})$-measurable.

Corollary

Let X, Y be metric spaces (or topological spaces) with Borel σ-algebras $\mathcal{B}_X$, $\mathcal{B}_Y$. If $f \in C^0(X,Y)$, then f is $(\mathcal{B}_X, \mathcal{B}_Y)$-measurable.

Proof

Let $\mathcal{O}_Y$ be the collection of open sets in Y, so $\langle \mathcal{O}_Y \rangle = \mathcal{B}_Y$. By the Prop., it is enough to check that $f^{-1}(U) \in \mathcal{B}_X$ for every $U \in \mathcal{O}_Y$. Since f is continuous, $U \in \mathcal{O}_Y \implies f^{-1}(U)$ is open in $X \implies f^{-1}(U) \in \mathcal{B}_X$.

10.4 Real-Valued Functions

Notation. In analysis we mainly care about functions $f : X \to \mathbb{R}$ (or $\mathbb{C}$), and we use the target σ-algebra $\mathcal{B}_{\mathbb{R}}$ (or $\mathcal{B}_{\mathbb{C}}$). We simply write $f \in \mathrm{Meas}(X)$ to mean that f is $(\mathcal{M}, \mathcal{B}_{\mathbb{R}})$-measurable (resp. $(\mathcal{M}, \mathcal{B}_{\mathbb{C}})$-measurable).

Proposition

Let $(X, \mathcal{M})$ be a measurable space and $f : X \to \mathbb{R}$. The following are equivalent:

ⓐ $f \in \mathrm{Meas}(X)$;
ⓑ $f^{-1}((a,\infty)) \in \mathcal{M}$ for all $a \in \mathbb{R}$;
ⓒ $f^{-1}([a,\infty)) \in \mathcal{M}$ for all $a \in \mathbb{R}$;
ⓓ $f^{-1}((-\infty,a)) \in \mathcal{M}$ for all $a \in \mathbb{R}$;
ⓔ $f^{-1}((-\infty,a]) \in \mathcal{M}$ for all $a \in \mathbb{R}$.

Proof

Each of the families of rays (a,∞), $[a,\infty)$, $(-\infty,a)$, $(-\infty,a]$ $(a \in \mathbb{R})$ generates $\mathcal{B}_{\mathbb{R}}$ (Sections 3–4). By the Prop. on generating sets, it is enough to check measurability on any one of them.

10.5 Products and Complex-Valued Functions

Measurability also behaves well with respect to products.

Proposition

Let $(X, \mathcal{M})$ be a measurable space and $\{(Y_\alpha, \mathcal{N}_\alpha)\}_{\alpha \in A}$ measurable spaces, with product

$$Y = \prod_{\alpha \in A} Y_\alpha, \qquad \mathcal{N} = \bigotimes_{\alpha \in A} \mathcal{N}_\alpha,$$

and coordinate maps $\pi_\alpha : Y \to Y_\alpha$. Then $f : X \to Y$ is $(\mathcal{M}, \mathcal{N})$-measurable iff $f_\alpha := \pi_\alpha \circ f : X \to Y_\alpha$ is $(\mathcal{M}, \mathcal{N}_\alpha)$-measurable for all $\alpha \in A$.

Proof

$(\Rightarrow)$ Each π_α is $(\mathcal{N}, \mathcal{N}_\alpha)$-measurable, since $\pi_\alpha^{-1}(E_\alpha) \in \mathcal{N}$ for $E_\alpha \in \mathcal{N}_\alpha$ by definition of $\mathcal{N}$. So $f_\alpha = \pi_\alpha \circ f$ is measurable, as a composition.

($\Leftarrow$) Suppose every f_α is measurable. For $\alpha \in A$ and $E_\alpha \in \mathcal{N}_\alpha$,

$$f^{-1}\big(\pi_\alpha^{-1}(E_\alpha)\big) = (\pi_\alpha \circ f)^{-1}(E_\alpha) = f_\alpha^{-1}(E_\alpha) \in \mathcal{M}.$$

The tube sets $\pi_\alpha^{-1}(E_\alpha)$ generate $\mathcal{N} = \bigotimes_\alpha \mathcal{N}_\alpha$, so by the Prop. on generating sets, f is $(\mathcal{M},\mathcal{N})$-measurable.

Corollary

Let $(X,\mathcal{M})$ be a measurable space. Then $f : X \to \mathbb{C}$ is measurable iff $\operatorname{Re} f$ and $\operatorname{Im} f$ are measurable.

Proof

Identify $\mathbb{C}$ with $\mathbb{R}^2$. Then $\mathcal{B}_{\mathbb{C}} = \mathcal{B}_{\mathbb{R}^2} = \mathcal{B}_{\mathbb{R}} \otimes \mathcal{B}_{\mathbb{R}}$ (Section 4), with $\operatorname{Re} f = \pi_1 \circ f$ and $\operatorname{Im} f = \pi_2 \circ f$. Apply the last Prop.

10.6 Algebraic Operations and Limits

So far: continuous maps are measurable, measurability can be checked on a generating set, and it behaves well with respect to products and compositions. It remains to show that it behaves well with respect to *algebraic operations* and *limits*. A supremum of real functions can be $+\infty$ (e.g. $f_j \equiv j$), so here we allow values in $\overline{\mathbb{R}} = [-\infty,\infty]$.

Definition

For functions $f_j : X \to \overline{\mathbb{R}}$, $j = 1,2,\dots$, the functions $\sup_j f_j$, $\inf_j f_j$, $\limsup_j f_j$ and $\liminf_j f_j$ are defined pointwise, by applying the operation to the sequence of numbers $f_j(x)$ at each $x \in X$:

$$\Big(\sup_j f_j\Big)(x) := \sup_j f_j(x), \qquad \Big(\limsup_{j\to\infty} f_j\Big)(x) := \limsup_{j\to\infty} f_j(x) = \inf_{k\geq 1} \sup_{j\geq k} f_j(x),$$

and likewise $(\inf_j f_j)(x) := \inf_j f_j(x)$ and $(\liminf_{j\to\infty} f_j)(x) := \sup_{k\geq 1} \inf_{j\geq k} f_j(x)$. All four take values in $\overline{\mathbb{R}}$.

Explanation. At a point x, $\sup_{j\geq k} f_j(x)$ decreases as k grows, since fewer terms are involved, so the infimum over k is also the limit as $k \to \infty$. So $\limsup_j f_j(x)$ is the value that the sequence $f_j(x)$ keeps coming back up to, its largest limit point, and it always exists in $\overline{\mathbb{R}}$; $\liminf_j f_j(x)$ is the smallest limit point. Always $\liminf_j f_j \leq \limsup_j f_j$, with equality at x exactly when $\lim_j f_j(x)$ exists in $\overline{\mathbb{R}}$, and then all three agree. E.g. for $f_j = (-1)^j$, $\limsup_j f_j = 1$ and $\liminf_j f_j = -1$. For indicator functions this matches Section 2: $\limsup_j \chi_{E_j} = \chi_{\limsup_j E_j}$, as both equal 1 exactly at the points lying in infinitely many E_j.

Note. Let $\mathcal{B}_{\overline{\mathbb{R}}} := \{E \subseteq \overline{\mathbb{R}} \mid E \cap \mathbb{R} \in \mathcal{B}_{\mathbb{R}}\}$. For $f : X \to \overline{\mathbb{R}}$, $f \in \operatorname{Meas}(X)$ means that f is $(\mathcal{M},\mathcal{B}_{\overline{\mathbb{R}}})$-measurable. Since $\mathcal{B}_{\overline{\mathbb{R}}}$ is generated by the rays $(a,\infty]$, $a \in \mathbb{R}$, and also by the rays $[-\infty,a)$, $a \in \mathbb{R}$, the Prop. on generating sets gives

$$f \in \operatorname{Meas}(X) \iff f^{-1}((a,\infty]) \in \mathcal{M} \ \ \forall a \in \mathbb{R} \iff f^{-1}([-\infty,a)) \in \mathcal{M} \ \ \forall a \in \mathbb{R}.$$

Proposition

Let $(X, \mathcal{M})$ be a measurable space.

ⓐ (Algebraic operations) If $f, g : X \to \mathbb{C}$ and $f, g \in \text{Meas}(X)$, then $f + g,\ f \cdot g \in \text{Meas}(X)$.
ⓑ (Limits) If $f_j : X \to \overline{\mathbb{R}}$ and $f_j \in \text{Meas}(X)$ for $j = 1, 2, \ldots$, then the functions $X \to \overline{\mathbb{R}}$

$$g_1 := \sup_j f_j, \qquad g_2 := \inf_j f_j, \qquad g_3 := \limsup_{j\to\infty} f_j, \qquad g_4 := \liminf_{j\to\infty} f_j$$

are in $\text{Meas}(X)$.

Proof

ⓐ Define $F : X \to \mathbb{C} \times \mathbb{C}$, $F(x) := (f(x), g(x))$, and $\varphi, \psi : \mathbb{C} \times \mathbb{C} \to \mathbb{C}$, $\varphi(z, w) := z + w$, $\psi(z, w) := zw$.

- F is $(\mathcal{M}, \mathcal{B}_{\mathbb{C}\times\mathbb{C}})$-measurable by the product result, since its coordinates f, g are measurable and $\mathcal{B}_{\mathbb{C}} \otimes \mathcal{B}_{\mathbb{C}} = \mathcal{B}_{\mathbb{C}\times\mathbb{C}}$ ($\mathbb{C}$ is separable; Section 4).
- $\varphi, \psi \in C^0(\mathbb{C} \times \mathbb{C}, \mathbb{C})$, so they are $(\mathcal{B}_{\mathbb{C}\times\mathbb{C}}, \mathcal{B}_{\mathbb{C}})$-measurable by the continuity result.

Hence $f + g = \varphi \circ F$ and $f \cdot g = \psi \circ F$ are measurable, as compositions.

ⓑ (sup) By the Note on $\mathcal{B}_{\overline{\mathbb{R}}}$, it is enough to show that $g_1^{-1}((a, \infty]) \in \mathcal{M}$ for all $a \in \mathbb{R}$. Now $\sup_j f_j(x) > a$ iff $f_j(x) > a$ for some j, so

$$g_1^{-1}((a, \infty]) = \bigcup_{j=1}^{\infty} f_j^{-1}((a, \infty]),$$

and $f_j^{-1}((a, \infty]) \in \mathcal{M}$ for all j by assumption, so $g_1^{-1}((a, \infty]) \in \mathcal{M}$ since $\mathcal{M}$ is a σ-algebra.

(inf) Similarly, $\inf_j f_j(x) < a$ iff $f_j(x) < a$ for some j, so

$$g_2^{-1}([-\infty, a)) = \bigcup_{j=1}^{\infty} f_j^{-1}([-\infty, a)),$$

and by the same argument $g_2^{-1}([-\infty, a)) \in \mathcal{M}$ for all $a \in \mathbb{R}$, i.e. $g_2 \in \text{Meas}(X)$.

(lim sup) Set $h_k := \sup_{j\geq k} f_j$. Then $h_k \in \text{Meas}(X)$ for all $k \geq 1$ by the sup case, and $g_3 = \limsup_j f_j = \inf_k h_k \in \text{Meas}(X)$ by the inf case.

(lim inf) Similarly, $\tilde{h}_k := \inf_{j\geq k} f_j \in \text{Meas}(X)$ by the inf case, and $g_4 = \liminf_j f_j = \sup_k \tilde{h}_k \in \text{Meas}(X)$ by the sup case.

Note. If $f(x) = \lim_j f_j(x)$ exists for all $x \in X$, then $f \in \text{Meas}(X)$, since $f = g_3 = g_4$ in that case.

Explanation. This is Requirement ②, and here measurable functions do better than continuous ones: $f_n(x) = x^n$ on $[0, 1]$ are continuous, but their pointwise limit (0 on $[0, 1)$, 1 at $x = 1$) is not; it is still Borel measurable.

Corollary

If $f, g : X \to \mathbb{R}$ and $f, g \in \mathrm{Meas}(X)$, then $\max(f,g),\ \min(f,g) \in \mathrm{Meas}(X)$ (apply ⓑ to $f_1 = f$, $f_j = g$ for $j \geq 2$).

Upshot. This notion of measurability of functions satisfies all our requirements. It remains to define a robust notion of integration for measurable functions.

11 Simple Functions and the Integral

11.1 Positive and Negative Parts

Definition

For $f : X \to \mathbb{R}$, the *positive part* and *negative part* of f are

$$f^+(x) := \max(f(x), 0), \qquad f^-(x) := \max(-f(x), 0).$$

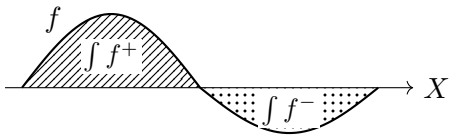
If $f \in \mathrm{Meas}(X)$, then $f^\pm \in \mathrm{Meas}(X)$ by the max Cor., since 0 and $-f$ are in $\mathrm{Meas}(X)$. So every $f \in \mathrm{Meas}(X)$ can be written as

$$f = f^+ - f^-, \qquad f^\pm \in \mathrm{Meas}(X), \quad f^\pm \geq 0. \tag{$*$}$$

The decomposition $(*)$ matters because we first define $\int_X f\,d\mu$ only for $f \in \mathrm{Meas}(X)$ with $f \geq 0$. Once $\int_X f^\pm\,d\mu$ are defined, $(*)$ lets us define

$$\int_X f\,d\mu := \int_X f^+\,d\mu - \int_X f^-\,d\mu$$

(as long as the two integrals are not both ∞). The sign requirement $f \geq 0$ is needed at first because of the approximation result by simple functions below.



Explanation. At each x at most one of $f^+(x)$, $f^-(x)$ is non-zero, so also $|f| = f^+ + f^-$.

11.2 Polar Decomposition

Every $f : X \to \mathbb{C}$ can be written as

$$f = (\operatorname{sgn} f)|f|, \qquad \text{where} \quad \operatorname{sgn} z := \begin{cases} z/|z|, & z \neq 0, \\ 0, & z = 0, \end{cases} \qquad z \in \mathbb{C}.$$

If $f \in \mathrm{Meas}(X)$, then $|f| \in \mathrm{Meas}(X)$ and $|f| \geq 0$: indeed $|f| = g \circ f$ with $g(z) := |z|$, and $g \in C^0(\mathbb{C}, \mathbb{R})$ is $(\mathcal{B}_\mathbb{C}, \mathcal{B}_\mathbb{R})$-measurable by the continuity Cor.

Alternatively, write a measurable $F : X \to \mathbb{C}$ as $F = \operatorname{Re} F + i \operatorname{Im} F$, where $\operatorname{Re} F, \operatorname{Im} F \in \operatorname{Meas}(X)$ are *real*-valued, and decompose further:

$$\operatorname{Re} F = (\operatorname{Re} F)^+ - (\operatorname{Re} F)^-, \qquad \operatorname{Im} F = (\operatorname{Im} F)^+ - (\operatorname{Im} F)^-.$$

So F is a combination of four non-negative measurable functions.

11.3 Simple Functions

Motivation. For a piecewise-constant $f : [0,1] \to \mathbb{R}$, i.e. $f = c_j$ on finitely many disjoint intervals E_j with union $[0,1]$, the Riemann integral is trivial: $\int_0^1 f\,dx = \sum_j c_j |E_j|$. Simple functions are the analogue with arbitrary measurable sets E_j.

Definition

For $E \subseteq X$, the *characteristic* (*indicator*) *function* of E is

$$\chi_E(x) := \begin{cases} 1, & x \in E, \\ 0, & x \notin E. \end{cases}$$

Note. $\chi_E \in \operatorname{Meas}(X) \iff E \in \mathcal{M}$, since $\chi_E^{-1}(B)$ is $\emptyset$, E, E^c or X, according to which of $0, 1$ lie in B.

Definition

Let $E_1, \ldots, E_n \in \mathcal{M}$ be disjoint with $\bigcup_{j=1}^n E_j = X$. A function $f : X \to \mathbb{C}$ of the form

$$f = \sum_{j=1}^n z_j \chi_{E_j}, \qquad z_j \in \mathbb{C},$$

is called *simple*, written $f \in \operatorname{Sim}(X)$.

Note. $f, g \in \operatorname{Sim}(X) \implies f + g,\ fg \in \operatorname{Sim}(X)$: if $f = \sum_j z_j \chi_{E_j}$ and $g = \sum_k w_k \chi_{F_k}$, then $f + g$ and fg are constant on each of the sets $E_j \cap F_k$, which are in $\mathcal{M}$, disjoint, and cover X.

Explanation. A simple function is just a measurable function with finitely many values: if f takes the distinct values $w_1, \ldots, w_m$, take $E_j := f^{-1}(\{w_j\})$.

11.4 The Approximation Theorem

The following *fundamental* result says that every $f \in \operatorname{Meas}(X)$ with $f \geq 0$ can be "well approximated" by simple functions. (This generalizes the approximation of continuous functions by piecewise-constant ones in the Riemann theory.)

Theorem (Approximation)

Let $(X, \mathcal{M})$ be a measurable space.

(a) Let $f : X \to [0, \infty]$, $f \in \mathrm{Meas}(X)$. Then there are $\varphi_1, \varphi_2, \ldots \in \mathrm{Sim}(X)$ such that
  (i) $0 \le \varphi_1 \le \varphi_2 \le \varphi_3 \le \cdots \le f$;
  (ii) $\varphi_n \to f$ pointwise.
  Moreover, $\varphi_n \to f$ uniformly on every set where f is bounded.

(b) Let $f : X \to \mathbb{C}$, $f \in \mathrm{Meas}(X)$. Then there are $\varphi_1, \varphi_2, \ldots \in \mathrm{Sim}(X)$ such that
  ① $0 \le |\varphi_1| \le |\varphi_2| \le \cdots \le |f|$;
  ② $\varphi_n \to f$ pointwise, and uniformly on every set where f is bounded.

Proof (a)

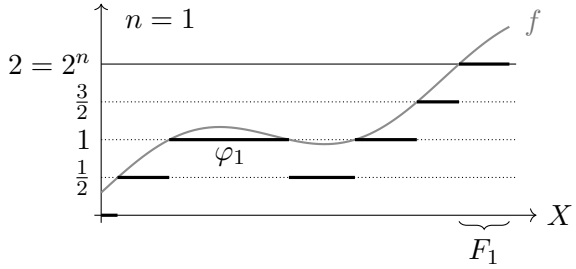
For $n = 1, 2, \ldots$ and $0 \le k \le 2^{2n} - 1$ set

$$E_n^k := f^{-1}\Big(\Big(\frac{k}{2^n}, \frac{k+1}{2^n}\Big]\Big), \qquad F_n := f^{-1}((2^n, \infty]),$$

and define

$$\varphi_n := \sum_{k=0}^{2^{2n}-1} \frac{k}{2^n} \chi_{E_n^k} + 2^n \chi_{F_n}.$$

The intervals $(k/2^n, (k+1)/2^n]$, $0 \le k \le 2^{2n} - 1$, partition $(0, 2^n]$. So the sets E_n^k, F_n and $f^{-1}(\{0\})$ are in $\mathcal{M}$ (as f is measurable), disjoint, and cover X. Hence $\varphi_n \in \mathrm{Sim}(X)$, with $\varphi_n = 0$ on $f^{-1}(\{0\})$.

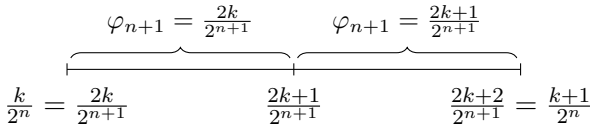


For $n = 2, 3, \ldots$ the dyadic scale in the range is refined.

Claim

$\varphi_n \le \varphi_{n+1}$ for all n.

To see this, split each dyadic interval of level n into two of level $n+1$:



Then

$$E_n^k = f^{-1}\Big(\Big(\frac{2k}{2^{n+1}}, \frac{2k+1}{2^{n+1}}\Big]\Big) \cup f^{-1}\Big(\Big(\frac{2k+1}{2^{n+1}}, \frac{2k+2}{2^{n+1}}\Big]\Big) = E_{n+1}^{2k} \cup E_{n+1}^{2k+1}.$$

On E_n^k we have $\varphi_n = \frac{k}{2^n} = \frac{2k}{2^{n+1}}$, while

$$\varphi_{n+1}\big|_{E_{n+1}^{2k}} = \frac{2k}{2^{n+1}}, \qquad \varphi_{n+1}\big|_{E_{n+1}^{2k+1}} = \frac{2k+1}{2^{n+1}} > \frac{2k}{2^{n+1}}.$$

So $\varphi_{n+1} \geq \varphi_n$ on each E_n^k. On F_n (where $f > 2^n$) we have $\varphi_n = 2^n$ and $\varphi_{n+1} \geq 2^n$: either $f > 2^{n+1}$ and $\varphi_{n+1} = 2^{n+1}$, or $f \in \left(\frac{k}{2^{n+1}}, \frac{k+1}{2^{n+1}}\right]$ for some $k \geq 2^{2n+1}$ and $\varphi_{n+1} = \frac{k}{2^{n+1}} \geq 2^n$. On $f^{-1}(\{0\})$ both vanish. Hence $\varphi_{n+1} \geq \varphi_n$, proving the claim.

Next, on E_n^k we have $\frac{k}{2^n} < f \leq \frac{k+1}{2^n}$, so $0 < f - \varphi_n \leq 2^{-n}$ there. On F_n, $\varphi_n = 2^n < f$, and on $f^{-1}(\{0\})$, $\varphi_n = 0 = f$. Hence $0 \leq \varphi_n \leq f$ everywhere, and since $\{f \leq 2^n\} = f^{-1}(\{0\}) \cup \bigcup_k E_n^k$,

$$0 \leq f - \varphi_n \leq 2^{-n} \qquad \text{on } \{f \leq 2^n\}. \tag{$\dagger$}$$

If $f(x) < \infty$, then $f(x) \leq 2^n$ for all large n, so $\varphi_n(x) \to f(x)$ by ($\dagger$). If $f(x) = \infty$, then $x \in F_n$ for all n, so $\varphi_n(x) = 2^n \to \infty = f(x)$. Thus $\varphi_n \nearrow f$ pointwise.

Finally, if $f \leq M < \infty$ on a set $E \subseteq X$, then $E \subseteq \{f \leq 2^n\}$ once $2^n \geq M$, so by ($\dagger$)

$$\sup_{x \in E} |f(x) - \varphi_n(x)| \leq 2^{-n} \longrightarrow 0,$$

i.e. $\varphi_n \to f$ uniformly on every set where f is bounded.

Proof (b)

Exercise. (*Hint:* apply (a) to $(\operatorname{Re} f)^{\pm}$ and $(\operatorname{Im} f)^{\pm}$.)

Explanation. Riemann sums cut the *domain* into intervals. Here the *range* is cut instead, into the dyadic intervals $(k2^{-n}, (k+1)2^{-n}]$, and on the set E_n^k where f lies in such an interval, φ_n is its lower endpoint.

11.5 Null Sets and Measurability

Sets of measure zero are irrelevant for measurability:

Proposition

Let $(X, \mathcal{M}, \mu)$ be a complete measure space.

① If $f \in \operatorname{Meas}(X)$ and $f = g$ μ-a.e., then $g \in \operatorname{Meas}(X)$.
② If $f_n \in \operatorname{Meas}(X)$ for all n and $f_n \to f$ μ-a.e., then $f \in \operatorname{Meas}(X)$.

Proof

Exercise. (*Hint:* if $f = g$ outside a null set N, then $f^{-1}(B)$ and $g^{-1}(B)$ differ only inside N; use completeness. For ②, apply ① to $\limsup_n f_n$.)

11.6 Integral of Simple Functions

Let $(X, \mathcal{M}, \mu)$ be a measure space. We first define $\int \varphi \, d\mu$ for simple $\varphi \geq 0$.

Notation. $L^+ := \{f : X \to [0, \infty] \mid f \in \operatorname{Meas}(X)\}$.

A function $\varphi \in L^+ \cap \operatorname{Sim}(X)$ (a non-negative simple function) can be written as

$$\varphi = \sum_{j=1}^{n} a_j \chi_{E_j}, \qquad E_j \in \mathcal{M} \text{ disjoint}, \quad \bigcup_{j=1}^{n} E_j = X, \quad a_j \geq 0.$$

Definition

For $\varphi \in L^+ \cap \mathrm{Sim}(X)$ as above,

$$\int_X \varphi \, d\mu := \sum_{j=1}^{n} a_j \, \mu(E_j),$$

with the convention $0 \cdot \infty = 0$. For $A \in \mathcal{M}$, set $\int_A \varphi \, d\mu := \int_X \varphi \chi_A \, d\mu$.

Remark. The value does not depend on how φ is written (used implicitly in the lecture). If also $\varphi = \sum_{k=1}^{m} b_k \chi_{F_k}$ as above, then $a_j = b_k$ whenever $E_j \cap F_k \neq \emptyset$, so $a_j \, \mu(E_j \cap F_k) = b_k \, \mu(E_j \cap F_k)$ for all j, k. Splitting E_j and F_k into the sets $E_j \cap F_k$ as in the proof of (b) below,

$$\sum_j a_j \, \mu(E_j) = \sum_{j,k} a_j \, \mu(E_j \cap F_k) = \sum_{j,k} b_k \, \mu(E_j \cap F_k) = \sum_k b_k \, \mu(F_k).$$

Explanation. For example, on $(\mathbb{R}, \mathcal{L}, m)$ the simple function $\chi_{\mathbb{Q}\cap[0,1]}$ (the limit function of Example 2 in Section 2) has $\int \chi_{\mathbb{Q}\cap[0,1]} \, dm = 1 \cdot m(\mathbb{Q} \cap [0,1]) + 0 \cdot m(\mathbb{R} \setminus (\mathbb{Q} \cap [0,1])) = 0$, where the second term is $0 \cdot \infty = 0$.

Next. Use this definition and the approximation theorem to define $\int_X f \, d\mu$ for general $f \in L^+$. First, some basic properties.

Proposition

Let $\varphi, \psi \in \mathrm{Sim}(X) \cap L^+$, and write $\int \varphi := \int_X \varphi \, d\mu$. Then:

(a) $c \geq 0 \implies \int c\varphi = c \int \varphi$;
(b) $\int (\varphi + \psi) = \int \varphi + \int \psi$;
(c) $\varphi \leq \psi \implies \int \varphi \leq \int \psi$;
(d) $\nu(A) := \int_A \varphi \, d\mu$, $A \in \mathcal{M}$, defines a measure ν on $\mathcal{M}$.

Proof

(a) $c\varphi = \sum_j c a_j \chi_{E_j}$, so $\int c\varphi = \sum_j c a_j \, \mu(E_j) = c \int \varphi$; for $c = 0$ both sides are 0, as $0 \cdot \infty = 0$.

(b) Write $\varphi = \sum_{j=1}^{n} a_j \chi_{E_j}$ and $\psi = \sum_{k=1}^{m} b_k \chi_{F_k}$ as above. Then

$$E_j = \bigcup_{k=1}^{m} (E_j \cap F_k), \qquad F_k = \bigcup_{j=1}^{n} (E_j \cap F_k)$$

are disjoint unions, since the F_k (resp. the E_j) are disjoint with union X. Also $\varphi + \psi = \sum_{j,k} (a_j + b_k) \chi_{E_j \cap F_k}$, where the sets $E_j \cap F_k$ are disjoint with union X. So by additivity of μ,

$$\begin{aligned} \int \varphi + \int \psi &= \sum_{j,k} a_j \, \mu(E_j \cap F_k) + \sum_{j,k} b_k \, \mu(E_j \cap F_k) \\ &= \sum_{j,k} (a_j + b_k) \, \mu(E_j \cap F_k) = \int (\varphi + \psi). \end{aligned}$$

ⓒ If $\varphi \le \psi$, then $a_j \le b_k$ whenever $E_j \cap F_k \neq \emptyset$; when $E_j \cap F_k = \emptyset$, the corresponding terms below vanish. So

$$\int \varphi = \sum_{j,k} a_j\, \mu(E_j \cap F_k) \le \sum_{j,k} b_k\, \mu(E_j \cap F_k) = \int \psi.$$

ⓓ For $A \in \mathcal{M}$, $\varphi\chi_A = \sum_j a_j \chi_{A\cap E_j} + 0 \cdot \chi_{A^c}$, where $A \cap E_1, \dots, A \cap E_n, A^c$ are disjoint with union X. So

$$\nu(A) = \sum_{j=1}^{n} a_j\, \mu(A \cap E_j),$$

and in particular $\nu(\emptyset) = 0$. Let $\{A_k\}_{k=1}^{\infty} \subseteq \mathcal{M}$ be disjoint and $A = \bigcup_k A_k$. By countable additivity of μ,

$$\nu(A) = \sum_{j=1}^{n} a_j \sum_{k=1}^{\infty} \mu(A_k \cap E_j) = \sum_{k=1}^{\infty}\sum_{j=1}^{n} a_j\, \mu(A_k \cap E_j) = \sum_{k=1}^{\infty} \nu(A_k).$$

So ν is countably additive, hence a measure.

12 The Monotone Convergence Theorem

12.1 The Integral of a Non-negative Function

Definition

Let $f \in L^+$. Then

$$\int_X f\, d\mu := \sup\left\{ \int_X \varphi\, d\mu \;\middle|\; \varphi \in \mathrm{Sim}(X),\ 0 \le \varphi \le f \right\},$$

and $\int_E f\, d\mu := \int_X f\chi_E\, d\mu$ for $E \in \mathcal{M}$.

Note. The set on the right is non-empty ($\varphi = 0$ is allowed, as are the φ_n of the approximation theorem), so the definition makes sense. For $\varphi \in \mathrm{Sim}(X) \cap L^+$ it agrees with the earlier definition, by ⓒ of the last Prop. Also, if $f, g \in L^+$ and $f \le g$, then $\int f \le \int g$ (monotonicity), since every simple φ with $0 \le \varphi \le f$ also satisfies $\varphi \le g$.

Explanation. Only approximation from below is used (there are no "upper sums"), so f may be unbounded and $\int f = \infty$ is allowed. The MCT below shows that $\int f = \lim_n \int \varphi_n$ for *any* simple $\varphi_n \nearrow f$, e.g. those of the approximation theorem, so a single increasing sequence computes $\int f$.

12.2 The Monotone Convergence Theorem

Theorem (MCT)

Let $(X, \mathcal{M}, \mu)$ be a measure space and $\{f_n\}_{n=1}^{\infty} \subseteq L^+$ such that

① $f_n \le f_{n+1}$ for all n ($f_n \nearrow$);
② $f = \lim_n f_n$ pointwise (so $f = \sup_n f_n \in L^+$).

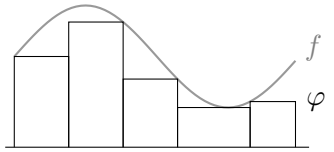
Then
$$\int_X f\,d\mu = \lim_{n\to\infty}\int_X f_n\,d\mu, \qquad \text{i.e.} \qquad \int_X \lim_n f_n\,d\mu = \lim_n \int_X f_n\,d\mu.$$

Warning. Do *not* apply MCT to functions that change sign.

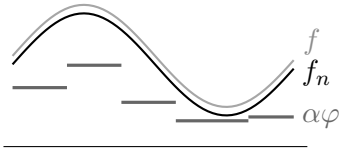
Proof

Since $f_n \le f_{n+1}$, monotonicity gives $\int_X f_n \le \int_X f_{n+1}$. So $\{\int_X f_n\}$ is increasing, and $L := \lim_n \int_X f_n\,d\mu$ exists in $[0,\infty]$. Since $f_n \le f$ for all n, monotonicity also gives $\int_X f_n \le \int_X f$ for all n, so
$$L \le \int_X f\,d\mu. \tag{$*$}$$
It remains to prove the other direction,
$$L \ge \int_X f\,d\mu. \tag{$**$}$$

Idea. Take a simple φ with $0 \le \varphi \le f$ and "squeeze" f_n between φ and f. Since $\varphi = f$ is possible at some points, first shrink φ to $\alpha\varphi$ with $0 < \alpha < 1$. Then $\alpha\varphi < f$ wherever $f > 0$, and $f_n \ge \alpha\varphi$ on more and more of X as n grows.



$0 \le \varphi \le f$



$\alpha\varphi \le f_n \le f$ (n large)

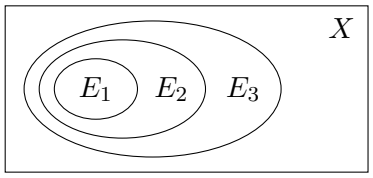
Fix $\alpha \in (0,1)$ and $\varphi \in \mathrm{Sim}(X)$ with $0 \le \varphi \le f$, and set
$$E_n := \{x \in X \mid f_n(x) \ge \alpha\varphi(x)\}.$$
Writing $\varphi = \sum_j a_j \chi_{G_j}$ with $G_j \in \mathcal{M}$ disjoint, $\bigcup_j G_j = X$ and $a_j \ge 0$, we get $E_n = \bigcup_j \left(G_j \cap f_n^{-1}([\alpha a_j, \infty])\right) \in \mathcal{M}$. Moreover:

(i) $E_n \subseteq E_{n+1}$, since $f_{n+1} \ge f_n$.

(ii) $\bigcup_{n=1}^{\infty} E_n = X$. Indeed, let $x \in X$. If $\varphi(x) = 0$, then $f_1(x) \ge 0 = \alpha\varphi(x)$, so $x \in E_1$. If $\varphi(x) > 0$, then $\alpha\varphi(x) < \varphi(x) \le f(x)$ since $\alpha < 1$; as $f_n(x) \nearrow f(x)$, there is N_x with $f_n(x) \ge \alpha\varphi(x)$ for all $n \ge N_x$, i.e. $x \in E_n$ for $n \ge N_x$.

Since $f_n \ge f_n\chi_{E_n} \ge \alpha\varphi\chi_{E_n}$ (as $f_n \ge \alpha\varphi$ on E_n), monotonicity, the definition of $\int_{E_n}$ and ⓐ give
$$\int_X f_n \ge \int_{E_n} f_n \ge \alpha \int_{E_n} \varphi. \tag{1}$$

By ⓓ, $\nu(A) := \int_A \varphi\,d\mu$ is a measure. So by (i), (ii) and continuity from below of ν,
$$\lim_{n\to\infty} \int_{E_n} \varphi = \int_X \varphi. \tag{2}$$



Letting $n \to \infty$ in (1) and using (2),

$$L \geq \alpha \int_X \varphi \qquad \text{for all } 0 < \alpha < 1$$

(also when $\int_X \varphi = \infty$, since $\alpha > 0$). Letting $\alpha \to 1$,

$$L \geq \int_X \varphi \qquad \text{for all } \varphi \in \mathrm{Sim}(X) \text{ with } 0 \leq \varphi \leq f. \tag{3}$$

The argument holds for *any* such φ, so taking the sup over φ in (3) gives $L \geq \int_X f\, d\mu$, which is $(**)$. With $(*)$, $L = \int_X f\, d\mu$.

Explanation. The hypothesis $f_n \geq 0$ cannot be dropped: on $(\mathbb{R}, \mathcal{L}, m)$ the functions $f_n := -\chi_{[n,\infty)}$ increase to 0 pointwise, but $\int f_n = -m([n,\infty)) = -\infty$ for all n, while $\int 0 = 0$.

12.3 MCT for Series

In analysis we want to interchange integrals and limits (MCT), and also integrals and sums.

Theorem (MCT + Summation)

Let $(X, \mathcal{M}, \mu)$ be a measure space, $\{f_n\}_{n=1}^{\infty} \subseteq L^+$, and

$$f := \sum_{n=1}^{\infty} f_n = \lim_{N\to\infty} S_N, \qquad S_N := \sum_{n=1}^{N} f_n$$

(the limit exists in $[0,\infty]$, since $f_n \geq 0$ makes S_N increase in N). Then

$$\int_X f\, d\mu = \sum_{n=1}^{\infty} \int_X f_n\, d\mu, \qquad \text{i.e.} \qquad \int_X \sum_{n=1}^{\infty} f_n\, d\mu = \sum_{n=1}^{\infty} \int_X f_n\, d\mu.$$

Proof

First consider two functions $f_1, f_2 \in L^+$. By the approximation theorem there are $\varphi_j, \psi_j \in \mathrm{Sim}(X) \cap L^+$ with

$$\varphi_j \nearrow f_1, \qquad \psi_j \nearrow f_2 \qquad (j \to \infty).$$

Then $\varphi_j + \psi_j \in \mathrm{Sim}(X) \cap L^+$ and $\varphi_j + \psi_j \nearrow f_1 + f_2$ (in particular $f_1 + f_2 \in L^+$, as a pointwise limit of measurable functions). By MCT,

$$\int (f_1 + f_2) = \int \lim_j (\varphi_j + \psi_j) = \lim_j \int (\varphi_j + \psi_j). \tag{$\dagger$}$$

Since φ_j, ψ_j are simple, $\int(\varphi_j + \psi_j) = \int \varphi_j + \int \psi_j$ by ⓑ. So by (†) and MCT for $\varphi_j \nearrow f_1$ and $\psi_j \nearrow f_2$,

$$\int (f_1 + f_2) = \lim_j \int \varphi_j + \lim_j \int \psi_j = \int f_1 + \int f_2.$$

By induction, for every finite N,

$$\int \sum_{n=1}^{N} f_n = \sum_{n=1}^{N} \int f_n. \tag{‡}$$

Now let $N \to \infty$ in (‡). Since $S_N \in L^+$ (by the two-function case and induction) and $S_N \nearrow f$, MCT gives $\lim_N \int S_N = \int f$. On the other hand, $\lim_N \sum_{n=1}^{N} \int f_n = \sum_{n=1}^{\infty} \int f_n$. Therefore

$$\int \sum_{n=1}^{\infty} f_n = \sum_{n=1}^{\infty} \int f_n.$$